- Contact with overhead power lines can give an electric shock, burn or **kill**.
- Injury can also occur when a person or object gets too close to a line and a flashover (arcing) occurs.
- Establish a safe zone under and around the overhead power lines before any work starts. Ensure no materials or machinery are stored nearby.
- It is not only high voltage cables that are dangerous. Voltages lower than 230 volts can kill people.

7.3 Underground services

7.3.1 Risks and costs of damage

Damage to gas pipes can happen even when their location is known (such as contact by an excavator bucket).

Damage to water pipes and telephone cables is often less evident but still potentially serious, affecting local businesses, schools, hospitals and emergency services that are dependent on those services.

If plant damages a fibre optic cable, this could result in a long section needing to be replaced, in most cases costing tens of thousands of pounds. In some cases, fibre optic cable strikes have cost up to £250,000.

All utility companies take cable strikes and damage very seriously. Not only because of the risks to the lives of their employees but also because of the financial costs and negative impacts on future business.

Some utility companies have problems with retaining liability insurance if they have a poor record of utility strikes. Multi-million pound organisations have gone out of business as a result of being unable to obtain liability insurance.

Most investigations into cable strikes reveal human error, where procedures for locating, identifying, supporting and working safely around services are not followed or shortcuts have been taken.

Such errors and omissions have resulted in many fatalities and life-altering injuries to individuals, and the dismissal of persons involved.

In the event of a gas leak, suspected gas leak or any other emergency relating to gas, immediately phone the National Grid gas emergency service on 0800 111 999.

In the event of an electrical emergency, phone the National Grid 24-hour electrical emergency service on 0800 404 090.

www

The guidance document *Avoiding danger from underground services* (HSG47) can be downloaded from the HSE website.

7.3.2 Types of underground services

The most commonly encountered underground services are gas, water, electricity and telecommunications, but drains and sewers will also need to be located as they are underground services too.

There are other services that may not be known about or detected other than through investigation before digging or excavation is started; these include cable television, hydraulics, private telecommunications and street lighting cables.

In some cases, depending on the location, it may be necessary to investigate for petroleum and fuel oils (large bore pipelines linking major installations – the majority of these pipelines are buried deep, although some may have reduced cover and be shallower in certain situations).

If work in the vicinity of gas transmission pipelines is being considered, the local gas company will be able to provide details of suitable procedures.

For further information on types of underground services refer to Chapter C03 Electrical safety and Chapter F01 Street works and road works.

7.3.3 Colour coding of underground services

There is national agreement between the utility groups for colour coding of services.

www

Full information can be obtained from the NJUG guidelines on the *Positioning and colour coding of underground utilities' apparatus*.

You will, however, need to bear in mind that large numbers of pipes and services have been laid over the last 100 years, in a mixture of materials and colours.

Note: *The table below identifies that water and electricity can use different coloured pipes.*

Recommended colour coding for underground utility apparatus	
Service(s)	**Duct, pipe or cable**
Electricity, water, oil, fuel pipelines and telecoms	Black PVC
Water	Blue PVC
Water and telecoms	Grey PVC
Electricity	Red PVC
Telecoms	Natural clay (terracotta), or grey, white, green, black or purple PVC
Gas	Yellow PVC

Note: *all plastic, polyethylene (PE), and polyvinylchloride pipes are shown above as PVC.*

Pipe or cable material used by the different services	
Pipe or cable	**Service(s)**
Cast iron	Gas and water
Steel	Gas and water
Braided steel	Electricity
Yellow steel	Gas
Copper	Water
Lead or lead covered	Electricity and water
Asbestos	Water
Hessian wrapped	Electricity

(The information in the table above is for guidance purposes only.)

Considerations when identifying underground services

- The coal authority has laid some yellow PVC electric cable.
- Black PVC must always be assumed to be live electricity until proved otherwise.
- All cast iron and steel must be assumed to be carrying gas until proved otherwise.
- Ducts may contain any one of the services, irrespective of type or colour of the duct.

7.4 Plan

7.4.1 Checking for underground services

Before carrying out any work, check with all public and private utilities for existence of services in the area that work is planned for and ensure these service routes are clearly marked on the plans.

The local electricity or gas company will give advice by telephone. British Telecom (BT) operates a similar type of service, and those wishing to make such enquiries should phone and select the option 'dial before you dig'.

Be careful when reading plans; reference points may have been moved, surfaces regraded (increasing or reducing the depths of cover), services moved without authority and not all connections may be shown.

On old brownfield sites there may be abandoned services and underground metallic items (such as tram lines and cast iron pipes); these can distort survey results, as can depth, certain ground conditions and concrete slabs.

Once routes have been identified, mark them with paint, tape or markers but **not** metal spikes, which might penetrate a cable or pipe.

Remember that the exact position will only be known when the underground service is uncovered.

Changes in colour of the surface material (sometimes known as tracks) may indicate where a service trench can be found. Absence of marker posts is not evidence that there are no underground services.

Shallow service

An electricity cable was damaged by the floor saw used to cut the road prior to excavating a new service trench. Although a cable-avoidance tool (CAT) survey had located the cable, it was discovered that it was directly under the asphalt sub-base, which was 270 mm thick.

Never assume that services have been installed at the recommended depth – they are often shallower. Never assume that a located service is the only one; there may be others adjacent to, above or underneath it.

Beware of services encased in concrete bases or structures, or in the concrete backing to kerbs.

Beware of services rising over obstructions, culverts, bridges, and so on. They are often much shallower in these locations.

Services leading to buildings and properties often also rise near to the point of entry to the building, due to the building foundations.

The presence of the following items indicates that underground services exist.

- Lamp posts (street lighting).
- Illuminated traffic signs.
- Telephone boxes.
- Concrete or steel inspection chamber covers.
- Hydrant and valve pit covers.
- Steel cabinets in the footway (set back) or in a verge.

Types of marker posts commonly found to indicate the location of underground services

The Utility Strike Avoidance Group (USAG) has a free online toolkit, *Best practice in avoiding underground services.*

The toolkit has been prepared by a working group made up of contractors, asset owners and utility service providers. The aim is to improve the risk management of work on and around underground services and to reduce the number and severity of utility strikes.

7.5 Locate

7.5.1 Using cable and pipe locators

Be aware that the CAT and signal generator should be used together, to provide the full picture. Equipment should be regularly serviced and calibrated, at intervals not exceeding 12 months.

All equipment of this type must have valid calibration certificates. Users should receive formal training in the make and model of the location equipment being used.

Recent developments in technology include depth read-outs, and some models include global positioning systems (GPSs) for accurately mapping the location of services.

Power frequency. A CAT will find most electricity cables whilst power is flowing through them, but it may not work in the following situations.

- If the cable has been disconnected.
- When the loading on a three-phase supply is evenly distributed.
- If the current is so small it is beyond the detection capability of the tool.
- If no current is flowing as the device is inactive (such as street lighting in the daytime).

Radio frequency can be used to detect electricity cables that have not been found by power detection, but its limitations can be geographical and it can also pick up other metal objects.

Radio frequency can be used to detect telecommunications cables of certain frequencies, but may not work on fibre-optic cables.

An example of a CAT

The **transmitter and receiver** (inductive or conductive) mode is the most suitable when there is no current in the services being sought.

Conventional cable locators will not find plastic pipes.

A **generator** (genny), when attached to an exposed part of a pipe or cable, will provide a signal for a CAT to track; it is important to continue to use the locator as the excavation progresses.

Metal detectors can detect hidden flat metal covers, joint boxes, and so on, but can miss cables or pipes. The deeper these objects are buried, the less chance of detection.

Ground-penetrating radar is a portable transmitter that can sweep the area of land where there are thought to be underground services.

The transmitter will display variations in the materials below the surface and can show where ground has been disturbed.

An operative properly trained in the use of radar equipment can detect the majority of underground services.

Whichever locating equipment you are using you must always compare your findings with existing plans to identify any differences between the supposed and actual locations.

Information on all cables and pipes detected should be recorded and passed to the principal designer for inclusion in the client's health and safety file.

Ensure that co-ordinates, lines and levels of newly installed services are included.

7.6 Dig – excavating

7.6.1 Permit to work and permit to dig

Where necessary, a formal permit-to-work system should be employed. A permit system will show the precautions that must be taken, what has been done and what is required. The permit to work also enables the person in charge to check that all the conditions have been met before work commences.

Some sites operate a system where a copy of the permit to dig is held by the worker in an armband. This allows site management and supervisors to monitor that excavation work is being undertaken under a permit.

7.6.2 Digging methods

Depending upon the potential hazards, it may be necessary to use a permit-to-work system before commencing any digging. On some sites a specific permit to dig system is used prior to breaking the ground.

*Deep excavation with sheet piling (**Note:** black sleeving on white cable indicating repair to strike damage)*

Hand digging near live services

Trial holes should be dug by hand to establish the exact location and depth of the service; **never** assume the service runs in a direct line or at the same depth between two trial holes.

Do not use power tools or excavators within 500 mm of services. Hand digging must be adopted using insulated tools (such as spades or shovels). Excavate alongside the service rather than directly above it.

Final exposure of the service by horizontal digging is recommended, as the force applied to hand tools can be controlled more effectively.

Power tools can be used to break paved surfaces, but be careful to avoid over-penetration as services near buildings may be much closer to the surface than the recommended minimum depth.

Do not use power tools directly over an indicated line of a cable unless it has been made dead or actions have been taken to ensure safety and avoid damage.

Where excavations are supported with shuttering or trench boxes, the sides must be designed to accommodate any services that pass through the excavation.

Services may need to be supported in these surroundings.

Never assume that a service is dead – always treat it as live until confirmed otherwise.

Checklist – prior to and during excavation

- Check with all utilities providers and landowners before starting work.
- Assume the presence of services when digging, even if nothing is shown on plans.
- Use detection devices and keep a close watch for signs of underground services (such as marker tape or tiles).
- Don't assume the services will be at the recommended minimum depths, they may be closer to the surface than normal, especially in the vicinity of works, structures or other services.
- Check that markers (such as plastic tape, tiles, slabs or battens) actually indicate the exact location of the underground service. Markers may have been displaced and will not necessarily be accurate.
- Some electric cables and water pipes look alike, as do some gas pipes and water pipes. Ensure each pipe is properly identified before starting work on them.
- Take additional care and use insulated spades and shovels when working near to services that can be easily damaged by a fork or a pick axe forced into the ground.
- Depending upon the risk assessment, those that are likely to encounter live services on a regular basis should wear flame resistant clothing, gloves and suitable eye protection for the task.
- Carefully lever out rocks, stones and boulders.
- Avoid over-penetration of the ground or surface with hand-held power tools as this is a common cause of accidents.
- If an excavator or digger has to be used near any service, the task will need to be specifically risk assessed to prevent accidental damage. Where possible, no-one should be near the digger bucket while it is digging.
- Ensure the excavator operator and others excavating are informed of the presence of suspected services.
- If the service is embedded in concrete or paving material, the owner may be able to de-energise it. Otherwise, make it safe or approve a safe system of work before it is broken out.
- Always assume closed, capped, sealed, loose or pot-ended services are live or charged, not dead or abandoned, until proved otherwise.
- Follow the guidelines and advice issued by the electricity, gas, water and telecommunication industries.
- Where possible, carry out the final exposure of underground services in a way that prevents any damage (such as using a vacuum excavator or compressed air lance).

For further information refer to Chapter D06 Excavations.

7.6.3 Piling and drilling

A safe system of work must be devised and implemented before any piling or drilling takes place. If it is possible that there are services nearby, surveys should be carried out and their position must be exposed by hand digging first.

www **For further information and guidance on recommended good practices and safety in piling operations, visit the Federation of Piling Specialists website.**

7.6.4 Exposure and protection

If a service in a trench is exposed, protect it with wood or other suitable materials to prevent any damage to the service. Services crossing trenches need support to avoid stress. Seek advice from the relevant utility company, if necessary. **Never** use services crossing in a trench as footholds, anchorage or climbing points.

www **For information on working adjacent to underground pipelines and the location of underground pipelines around the UK visit the Linewatch and Linesearch websites.**

7.6.5 Backfilling

When backfilling a trench **do not** place hard core, surplus concrete, rock, rubble and flint onto a service pipe or cable, as it may damage it. **Do not** use wet spoil or any perishable materials for backfill. Use suitable material and ensure it is compacted with care to avoid shocking the service pipe or cable. Warning tape should then be placed approximately 300 mm above the service. If gas service pipes have been exposed, advice on backfilling should be obtained from the local gas company.

7.6.6 Emergency works

Emergency works often mean there is no time to contact the utility companies. However, this work can be carried out safely if:

- the area is marked out carefully
- detectors are used correctly
- trial holes are dug by hand
- the practice of safe digging is followed.

7.6.7 Working in the roadway

Only licensed utility companies may carry out work in the footway, carriageway or any verge. When working in the roadway, ensure compliance with the New Roads and Street Works Act 1991.

For further information refer to Chapter F01 Street works and road works.

7.7 Damage to underground services

Damage to underground services **must** be reported to the owner and/or occupier immediately. When such damage causes an emergency situation, call the **police**, **fire** and/or **ambulance** services as necessary.

7.7.1 Gas

In the event of a gas leak, suspected gas leak or any other emergency relating to gas, immediately phone the National Grid gas emergency service on 0800 111 999. This and all other emergency contact numbers and details should be readily accessible on site.

If a gas leak is suspected, contact the police, fire brigade and gas supply company immediately. If there is a dangerous situation and the emergency services have not arrived, try to evacuate the immediate area including, if necessary, the occupants of nearby properties to an upwind position.

As far as possible, prevent anyone from smoking and keep traffic clear of the area.

If the gas escape catches fire, do not attempt to extinguish the flames.

The normal minimum depth of cover for gas mains operating in the low and medium pressure ranges is:

- 600 mm in footways or verges
- 750 mm in carriageways.

These figures may vary since each gas company can have its own standards for certain situations.

Where service pipes serve as connections from mains services that run from the road as a supply to a building, they may be shallower, especially when near to any property or structure.

7.7.2 Electricity

In the event of an electrical emergency, phone the National Grid 24-hour electrical emergency service on 0800 404 090.

Avoid contact with any damaged cable or apparatus. If operating a machine, do not attempt to disentangle any equipment. If safe and if possible, jump clear of the machine, ensuring that contact is not made with the vehicle and ground at the same time.

Do not return to the machine. If this is not possible, stay exactly where you are. Shout for help. As far as possible, do not touch any metallic part of the vehicle, such as steel parts of the cab or door. If possible, inform the electricity company or ask someone else to. Keep people away.

The depth at which electricity cables or ducts are usually installed in the ground is decided by the need to avoid undue interference or damage. Depending on the type of cable and the power that it may be carrying, the depth of cover may vary from 450–900 mm.

However, existing services can be found at any depth. It is not uncommon for them to have as little as 50 mm cover. In all cases where the depth of cover is likely to increase or decrease, the service owner must be consulted.

7.7.3 Other services

Leave a damaged service well alone and inform the owner.

Some cables are automatically re-energised by the local sub-station after a short time following the supply tripping out due to damage. Do not assume that a damaged cable will remain dead.

7.8 Overhead services

Every year people are killed or seriously injured when they come into contact with overhead electricity power lines. Incidents occur when working activities are not properly planned and result in contact with power lines (for example, contact with tipping trailers, cranes, scaffold tubes and ladders).

If there is contact with, or even if a piece of equipment gets too close to a power line, the electricity can be conducted to earth, which can cause fire, an explosion and shock or burns to anyone touching the machine or equipment. Overhead lines can be difficult to spot, particularly in foggy or dull conditions. Often, people just fail to look up.

Electricity supplies above 33,000 volts are usually routed overhead. Supplies below this voltage may be either overhead or underground. There is a legal minimum height above ground level for overhead power lines that varies according to voltage carried, shown below.

Voltage carried	400 kV	275 kV	132 kV	33 kV-low voltage
Legal minimum height above ground level	7.3 m	7.0 m	6.7 m	5.2 m, except for roads where the minimum is 5.8 m

The law requires that work may only be carried out in close proximity to live overhead lines when there is no alternative, and only when the risks are acceptable and can be properly controlled.

Where work cannot be avoided, consult the local electricity company or DNO **before** any work is started; a safe system of work must be planned and implemented. Power lines should be isolated and made **dead** or suitable precautions taken to prevent danger before any work takes place. There may be other suppliers who have to be notified (such as Local Authorities, National Grid and other electricity companies). Where necessary, wait for the supplier to isolate or re-route the cable to enable the work to take place.

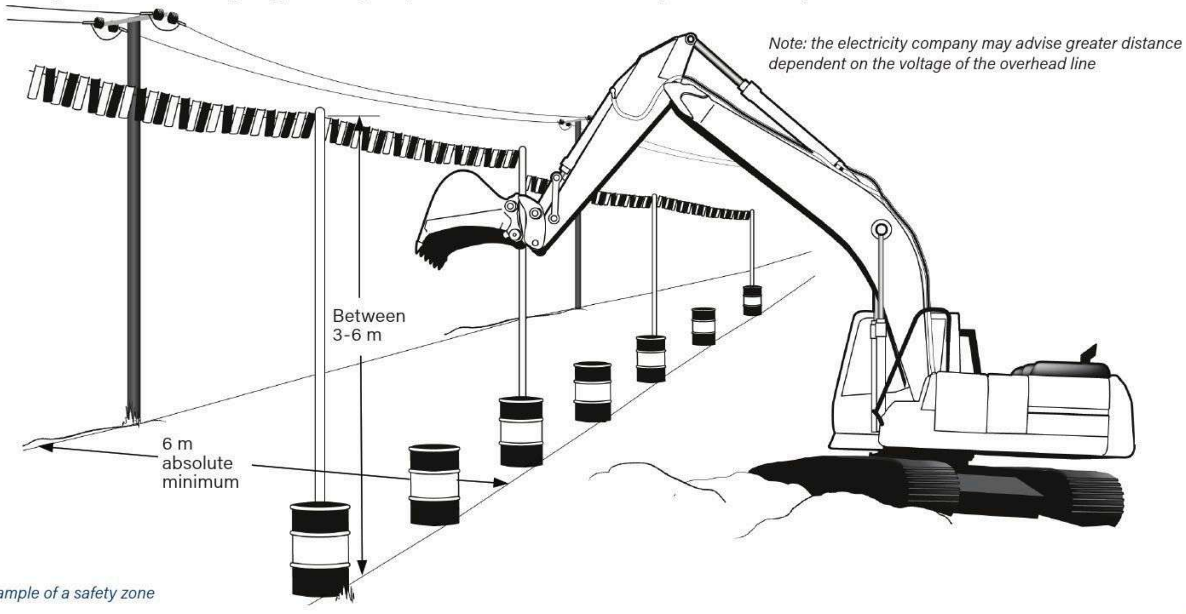
Take practical steps to prevent danger from any live cable or apparatus. This should include the placing of substantial and highly visible barriers. A barrier is only required one side if access is only possible from one side of the overhead power line (i.e. at the edge of the site), but if the line crosses the site, barriers will need to be placed on both sides. If there is a danger to people with scaffold poles or other conducting objects then the barriers should exclude people and mobile plant.

Electricity travels at the speed of light – more than 186,000 miles per second. The people you are responsible for don't.

7.8.1 Working near but not underneath overhead lines and the use of barriers

Where there will be no work or passage of machinery or equipment under the power lines, ground level barriers can be erected to establish a **safety zone**. This area should not be used to store materials or machinery. Suitable barriers can be constructed out of large, steel drums filled with rubble, concrete blocks, wire fence, earthed at both ends, or earth banks marked with posts.

- If steel drums are used, highlight them by painting them with, for example, red and white horizontal stripes.
- Make sure the barriers can be seen at night, perhaps by using white or fluorescent paint or attaching reflective strips.
- The safety zone should extend at least 6 m horizontally from the nearest wire or apparatus on either side of the overhead line. You may need to increase this on the advice of the line owner, or to allow for the possibility of a jib or other moving part encroaching into the safety zone. The electricity supplier will give specific, on-site advice on the position of safety zones.



Example of a safety zone

Where plant (such as a crane) is operating in the area, additional high-level indication should be erected to warn operators. A line of coloured, plastic flags or bunting, mounted 3-6 m above ground level over the barriers, is suitable.

Extreme caution must be exercised when erecting bunting and flags to avoid contact or arcing with the wires. The electricity supplier will give specific, on-site advice on the positioning of high level indicators.

It must be assumed that all overhead lines and cables are live unless advised otherwise by the electricity company.

7.8.2 Safe working when passing underneath overhead lines

If equipment or machinery capable of breaching the safety zone has to pass underneath the overhead line, you will need to create a passageway through the barriers. In this situation you should ensure the following.

- The number of passageways is kept to a minimum.
- The route of the passageway is defined using fences, and goalposts are erected at each end to act as gateways using a rigid, non-conducting material (such as timber or plastic pipe) highlighted with, for example, red and white stripes.
- If a passageway is too wide to be spanned by a rigid, non-conducting goalpost, you may have to use tensioned steel wire, earthed at each end, or plastic ropes with bunting attached. These should be positioned further away from the overhead line to prevent them being stretched and the safety clearances being reduced by plant moving towards the line.
- The surface of the passageway is levelled, formed up and well maintained to prevent undue tilting or bouncing of the equipment.
- Warning notices are displayed at either side of the passageway, on or near the goalposts and on approaches to the crossing, giving the crossbar clearance height and instructing drivers to lower jibs, booms, tipper bodies, and so on, and to keep below this height while crossing.
- Notices and crossbars are illuminated at night or in poor weather, as required, to make sure they are visible.
- The barriers and goalposts are maintained.

Ensure arrangements have been made for the passage of tall plant at specific times where overhead power lines have been made dead.

Working on or near live conductors

Regulation 14 of the Electricity at Work Regulations for work on or near live conductors states:

- no person shall be engaged in any work activity on or so near any live conductor (other than one suitably covered with insulating material so as to prevent danger) that danger may arise unless:
  - it is unreasonable in all circumstances for it to be dead
  - it is reasonable in all circumstances for a person to work on or near it while it is live
  - suitable precautions (including where necessary the provision of suitable protective equipment) are taken to prevent injury.

7.8.3 Working underneath overhead lines

Exclusion zones should be set up around the line and any other equipment that may be fitted to the pole or pylon. If you cannot avoid transitory or short duration, ground-level work where there is a risk of contact from, for example, the upward movement of cranes or people carrying tools and equipment, you should carefully assess the risks and precautionary measures.

The minimum extent of these zones varies according to the voltage of the line, as follows.

- 1 m from low-voltage lines.
- 3 m from 11 kV and 33 kV lines.
- 6 m from 132 kV lines.
- 7 m from 275 kV and 400 kV lines.

For detailed advice about the line voltage or use of exclusion zones you should always consult the owner of the overhead line.

- Arrange for the work to be directly supervised by someone who is familiar with the risks and can make sure that the required safety precautions are observed.
- Make sure that workers, including any contractors, understand the risks and are provided with instructions about the risk prevention measures.
- Poles and hand tools should not be able to encroach within these zones. Allow for uncertainty in measuring the distances and for the possibility of unexpected movement of the equipment due, for example, to wind conditions.
- Carry long objects horizontally and close to the ground and position vehicles so that no part can reach into the exclusion zone, even when fully extended. Machinery (such as cranes and excavators) should be modified by adding physical restraints to prevent them reaching into the exclusion zone.

- Insulating guards and/or proximity warning devices fitted to the plant without other safety precautions are not considered adequate protection on their own.
- Machinery (such as cranes and excavators) should be modified by adding physical restraints to prevent them reaching into the exclusion zone.

Work should not take place close to or under an overhead line during darkness or poor visibility conditions. Dazzle from portable or vehicle lighting can obscure rather than illuminate power lines.

Electricity can kill. The correct information, instruction, training and supervision can help to keep workers, and others, alive.

Worker receives fatal electric shock from overhead power line

A trainee scaffolder received a fatal electric shock whilst handling a 6.4 m long scaffold tube that came into contact with an 11,000 volt overhead power line. No site-specific risk assessment had been carried out, only generic ones at the office.

The company employing the worker was prosecuted under the Health and Safety at Work etc. Act 1974, Section 2(1), and fined.

The company should have:

- closely supervised the trainee until he was able to demonstrate competence in a wide range of work situations
- undertaken a risk assessment for that site and informed all workers of the outcome
- taken action to reduce the risks from contact with overhead power lines (for example, requesting that the electricity distributor made the overhead power line dead during the work).

For further information refer to the HSE guidance note *Avoiding danger from overhead power lines* (GS6).

CONTENTS

Confined spaces

08

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.

Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

Every entry into a confined space is potentially hazardous. Accidents in confined spaces cause deaths at work, killing several people each year in a wide range of industries, including construction.

Some confined spaces are easy to identify (such as closed tanks, vessels and sewers). Others are less obvious but may be equally dangerous, including basement-level boiler rooms, toilets, lofts and attics, as well as open-topped tanks, vats, silos or other structures and excavations that become confined spaces during their construction, installation or manufacture.

8.1 Introduction

A confined space is a place which is both substantially or largely enclosed (though not always entirely), and where serious injury can occur from hazardous substances or conditions (such as lack of oxygen) within that space or nearby. It may be small and restrictive for the worker (like a loft space or inspection chamber) or it could be a far larger space (such as a basement or site storage silo with a capacity of hundreds of cubic metres).

Accidents in confined spaces are caused by a combination of factors arising from a lack of awareness, inadequate supervision and a lack of training. The situation is often made worse by heroic but ill-conceived rescue attempts, founded on insufficient planning and knowledge, which may lead to multiple fatalities. It is essential, therefore, to be able to identify confined spaces and the hazards associated with entering and working in them.

A *confined space* is any place, including any chamber, tank, vat, silo, pit, trench, pipe, sewer, flue, well or other similar space in which, by virtue of its enclosed nature, there arises a reasonably foreseeable *specified risk*.

In every situation, the planning of the work must first consider what measures could be taken to enable the work to be carried out properly without the need to enter the confined space.

If entry to a confined space is required the work must be properly planned to avoid any of the following specified risks.

- Injury to any person arising from a fire or explosion.
- Loss of consciousness of any person at work arising from an increase in body temperature.
- Loss of consciousness or asphyxiation of any person at work arising from gas, fumes, vapour or the lack of oxygen.
- Drowning of any person at work arising from an increase in the level of liquid.
- Asphyxiation of any person at work arising from a free-flowing solid or the inability to reach a respirable environment due to entrapment by a free flowing solid.

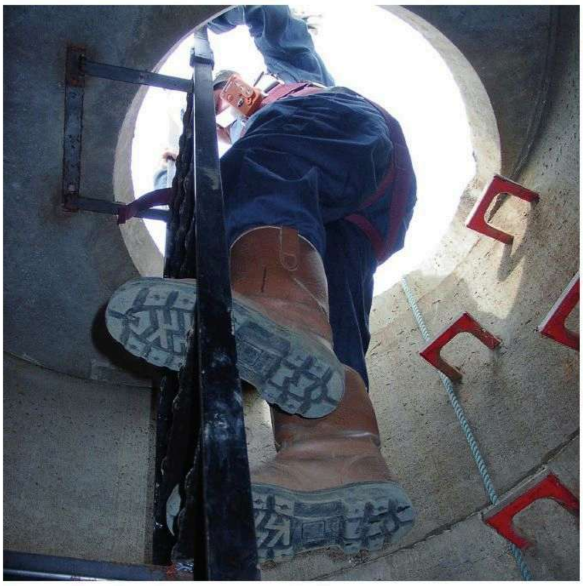
Places not usually considered to be confined spaces may become confined spaces because of a change in the condition inside or a change in the degree of enclosure or confinement.

Application of the Confined Spaces Regulations 1997 in any of these places will depend on the presence of a reasonably foreseeable risk of serious injury from a hazardous substance or condition within the space or nearby.

Below are some examples of confined spaces that may be encountered during construction work.

- Cellars, small bathrooms or an inadequately ventilated basement room.
- Boilers, boiler flues or chimneys.
- Inspection chambers, sewers, drains, excavations or deep trenches.
- Ceiling voids or ducts.
- Caissons or cofferdams.
- Loft spaces or plant rooms.
- Any room or enclosed space with poor ventilation, as these can become a confined space (for example, paint spraying in a room or power floating in a large hanger using petrol-powered equipment).

There are many more examples of confined spaces and the hazards will vary with the location, the type of work being carried out and the equipment or substances used.



Care must be taken when entering a confined space

08

8.2 Important points

- Working in confined spaces has the potential to be hazardous unless the appropriate controls are put in place.
- Many people have died as a result of work in confined spaces not being adequately planned, organised or safely carried out; many of them were would-be rescuers.
- Confined spaces are not just sewers and ducts; under the regulations many other work areas could also be classified as confined spaces.
- Always consider options to carry out the work without anyone having to enter a confined space.

If working in a confined space cannot be avoided:

- A risk assessment must be carried out for all work in a confined space.
- Where the findings of the risk assessment reveal significant risks to health or safety, they should be recorded in a method statement and safe system of work and included in all briefings.
- Entry to a confined space should be controlled by a permit to work and, where considered necessary, a separate permit to enter.
- Any plan of the work must consider the method of communication from inside to outside, raising the alarm and safely rescuing the people in the confined space should the situation become unsafe.
- Any training may need to be specific for the type of confined space (for example, a sewer entry course may not be appropriate for someone who has to work in a hot roof space). Training should extend not only to those entering confined spaces but also to managers, supervisors and emergency personnel.
- Where possible, the confined space should be certified as safe to work in without the need for respiratory protective equipment (RPE). If the risk assessment identifies the need for RPE, it must be suitable for the hazards present and compatible with other personal protective equipment (PPE). Wearing RPE and PPE can result in an increase in respiration and heat stress, and preventative measures may be required to reduce the risks of heat stress and dehydration.

Unplanned or poorly planned work in confined spaces has the potential to kill – reading and understanding this chapter is not a substitute for adequate training and experience.

Take great care identifying what is and what could be a confined space. Many confined space accidents occur in work areas that have simply not been identified as potential confined spaces.

8.3 Hazards in confined spaces

8.3.1 Oxygen deprivation and suffocation

The air that we breathe contains around 21% oxygen and, at that level, people can work without difficulty. A falling level of oxygen will create an increasingly serious situation if breathing apparatus is not worn. Generally, the symptoms referred to in the table *(below)* are experienced at the corresponding level of oxygen depletion.

Oxygen deprivation may be the result of any of the following.

- The displacement of oxygen by gas leaking in from elsewhere, or the deliberate introduction of purge gas (for example, nitrogen or argon).
- The displacement of oxygen by a naturally occurring gas (such as methane).
- Oxidisation, rusting or bacterial growth using up the oxygen in the air.
- Oxygen being consumed by people working and breathing, or by any process of combustion.
- Welding and other hot works.
- The discharge of a fire extinguisher containing carbon dioxide or other asphyxiating gas.
- Paint spraying or using chemicals for cleaning.

Oxygen depletion	Symptoms
19%	Tiredness (this is the normal acceptable minimum level for working).
17%	Judgement (decision making) is affected.
12%	Respiration is affected and fatigue experienced.
10%	Light-headedness and increasing breathing difficulties.
8%	Nausea and possible collapse.
6%	Respiration stops and death follows within minutes.

8.3.2 Hazards of excess oxygen

An oxygen-enriched atmosphere is a major hazard. Organic materials (such as oil and wood) become highly combustible, and ordinary materials (such as paper and clothing) will burn with exceptional ferocity.

An increase of only 4% oxygen is sufficient to create a hazard and this may occur inadvertently. In oxyacetylene and oxypropane processes, sometimes not all of the oxygen supplied to a cutting torch is consumed. Some may be released, increasing the atmospheric oxygen above the normal 21%.

The oxygen enrichment of the atmosphere in a confined space also results from the practice of using oxygen to sweeten or enrich the atmosphere when it has become oppressive, stale, hot, fume-filled or otherwise unpleasant. This is a dangerous practice and must be prohibited.

Another way in which the atmosphere may become oxygen-enriched is through leakage from torches or hoses that may go unnoticed (such as during meal breaks or overnight). For this reason, they should be removed and properly isolated at each breaktime. The deliberate kinking or nipping of an oxygen hose while changing a torch does not cut off the supply completely and can result in the release of substantial quantities of oxygen.

If excess oxygen is discovered, the space must be quickly evacuated and ventilated until normal levels of oxygen are regained.

8.3.3 Toxic atmospheres

However much oxygen is present in the atmosphere, if there is also a toxic gas present in sufficient quantity it will create a hazard.

Below are some of the many toxic gases that may be encountered.

- Hydrogen sulphide, usually from sewage or decaying vegetation, which can smell like rotten eggs.
- Carbon monoxide, from internal combustion engines, or any incomplete combustion, especially of liquefied petroleum gases (LPGs).
- Carbon dioxide, from any fermentation, or being naturally found in soil and rocks, or coming from the combustion of LPG.
- Fumes and vapours, from chemicals such as ammonia, chlorine, sodium, and from petrol and solvents.

Whenever a toxic gas (or any gas, fume or vapour that may be hazardous to health) is thought to be (or known to be) present, then an assessment of the risk to health must be made under the provisions of the **Control of Substances Hazardous to Health Regulations 2002 (COSHH Regulations)** and the appropriate control measures must be put into place to eliminate the hazard or control the risks.

Petrol and diesel engines create carbon monoxide, which is an extremely toxic gas.

Liquid petroleum gas-powered engines create an excess of carbon dioxide, which is a suffocating hazard.

The use of any form of internal combustion engine within a confined space must be prohibited, unless a specifically dedicated exhaust extraction system is in operation.

Carbon dioxide and carbon monoxide are colourless, odourless and tasteless. Both can cause loss of consciousness and death in a very short time. Acute exposure to high concentrations of hydrogen sulphide can result in collapse (knockdown), respiratory paralysis, cyanosis, convulsions, coma, cardiac arrhythmias and death within minutes.

8.3.4 Flammable and explosive atmospheres

Some gases and vapours need only be present in small quantities to create a hazard. Examples of major sources of explosive and flammable hazards are shown below.

- Petrol or LPG, propane, butane and acetylene. These are explosive in the range of 2% upwards in air. The hazard is normally created by a spill or leak.
- Methane and hydrogen sulphide, which are naturally evolved from sewage or decaying organic matter. These are explosive in the range of 5-15% in air for methane and 4-44% in air for hydrogen sulphide.
- Solvents, acetone, toluene, white spirit, alcohol, benzene, thinners, and so on. These are explosive in the range of 2% upwards in air. The hazard generally results from process plants and/or spillage.
- Hydrogen and other gases that are created from processes such as battery charging.

In an explosive or flammable atmosphere, a toxic or suffocating hazard may also exist.

8.3.5 Hostile environments

Apart from the hazards dealt with above, other dangers may arise from the use of electrical and mechanical equipment, from chemicals, process gases and liquids, dust, paint fumes, welding and cutting fumes.

Extremes of heat and cold can have adverse effects and may be intensified in a confined space. Consideration must be given to the timing of what would otherwise be considered standard work. During hot weather, roof spaces and other types of confined spaces may reach temperatures that will lead to a dangerous increase in body temperature.

If work cannot be planned to avoid this (for example, by starting early) then physical measures (such as cooling and reducing the time spent working in the confined space) must be introduced following an assessment by a competent person.

www **For further information and guidance on heat stress in the workplace, visit the HSE website.**

Further dangers exist in the sheer difficulty of getting into or out of, and working in, a confined space. The potential hazards of an inrush of water, gas or sludge due to a failure of walls or barriers, or leakage from valves, flanges or blanks, must all be considered at the risk assessment stage.

8.4 Confined Spaces Regulations 1997

These require employers to plan work so that entry to confined spaces is avoided so far as is reasonably practicable (for example, by doing the work from outside). They also require a safe system of work to be developed and implemented if entry to a confined space is unavoidable, and adequate emergency arrangements, which will also safeguard rescuers, to be put in place before work starts.

Duties to comply with the regulations are placed on:

- employers in respect of work carried out by their own employees and work carried out by any person (for example, a contractor) insofar as work is, to any extent, under the employer's control
- the self-employed in respect of work carried out by any other person insofar as work is, to any extent, under the control of the
  self-employed person.

The main duty under Regulation 4 is a complete prohibition of any person entering a confined space to carry out any work for any purpose whatsoever, where it is reasonably practicable to carry out the work by any other means.

Depending upon the work activity, any space can become a confined space, including lofts, lifts or inspection pits, service risers and unventilated rooms, even if they are large spaces

If entry into a confined space is necessary then a risk assessment by a competent person, under the **Management of Health and Safety at Work Regulations 1999**, is required. If your company does not have a competent person employed they must engage the services of a competent person from outside the company. The outcome of the risk assessment will then provide the basis for the development of a full and effective safe system of work, including rescue arrangements.

These regulations are supported and explained by an Approved Code of Practice (ACoP) and guidance notes. The competent person responsible for carrying out the risk assessment, or otherwise planning work in a confined space, should be familiar with all aspects of the regulations, including the ACoP and guidance material. If you follow the advice in the ACoP you will be doing enough to comply with the law in respect of those specific matters on which the Code gives advice.

For the definition of *reasonably practicable* refer to Chapter A01 Health and safety law.

www **The complete regulations and the ACoP and guidance notes, which includes a flowchart to help in the decision-making process, can be downloaded for free from the HSE website.**

8.5 Information, instruction and training

The need for comprehensive training prior to any involvement in confined space working cannot be stressed too highly. The number of deaths in confined spaces in recent years is testimony to the extreme hazards that can be present in confined spaces.

The information, instruction and training given to employees must enable them to carry out work safely and without risks to their health. The extent of training needed will vary according to circumstances and the type of space being entered. Any entry into a confined space requiring the use of breathing apparatus would require a full breathing apparatus and rescue training course.

However, training to enter a bund around a large diesel tank where the risks are less significant (such as fumes and possible drowning in diesel) would not require such an intensive course, and indeed adopting the use of breathing apparatus in this instance may be entirely inappropriate. Training should involve demonstrations and practical exercises. It is important that trainees are familiar with both equipment and procedures before working for the first time in confined spaces.

Practical refresher training should be organised and available. The frequency with which refresher training is provided will depend upon the length of time since the type of work was last carried out, or if there have been changes to methods of work, safety procedures or equipment.

The training needs for confined space working should be considered for each of the four categories of employee.

- Supervisors.
- Employees entering confined spaces.
- People employed as attendants outside of confined spaces.
- Rescue teams.

Some of the roles identified may be carried out by the same person.

There is currently no legal requirement to have obtained a specific standard of training prior to working in a confined space. However, there has been discussion between the utility companies, who are the clients for much of the sewer-entry work, with regard to recognising each other's training standards.

No person should enter a confined space unless they are trained and competent to do so safely.

Tunnelling offers more specific challenges. For the latest updates on training standards visit the Pipe Jacking Association and the British Tunnelling Society websites.

8.6 Safe working

If you cannot avoid entry into a confined space a safe system of work must be implemented. This can be achieved by the use of a permit to work system, in which each step is planned and all foreseeable hazards are taken into account. Such a system, backed up by adequate rescue facilities, should enable work to be carried out safely.

At the planning stage it will be necessary for the following information to be determined.

- Whether an entry into the confined space is required, or whether there is an alternative method of doing the work.
- If an entry is necessary, whether it can be carried out without the use of breathing apparatus.
- Whether the entry must be made with the use of breathing apparatus.

It should be emphasised that entry into a confined space using breathing apparatus should not be made routinely or undertaken as a matter of convenience, where the use of isolation and mechanical or forced ventilation would achieve a safe atmosphere.

If it is decided that the work can be done without anyone entering the confined space, provided that a safe system of work exists and the confined space has been isolated from potential sources of hazard, the work can proceed.

It is important to avoid systems or plant being re-energised while work is proceeding and everyone involved should be advised accordingly.

Once it has been decided that people must enter a confined space, a preliminary meeting should be held with all concerned, and effective lines of authority and communication established to minimise any risk of subsequent misunderstanding.

The exact routine to be followed will vary, depending on the type of confined space to be entered. The provisions and precautions required for entry into a large, empty surface water tank will obviously be different from those needed for entry into a narrow service duct containing pipes and valves, but the fundamental principle of a safe system of work applies to all cases.

The risk assessment, as mentioned previously, will have identified many of the above points and should be used as the basis for developing the safe system of work.

It is stressed that all PPE in general, and RPE in particular, must have been specified by a competent person who fully understands the circumstances surrounding its use, and that the persons wearing it are trained and competent in its use.

Employers must put in place adequate emergency arrangements before the work starts. No person must enter a confined space until suitable and sufficient procedures are in place to facilitate a rescue in the event of an emergency. Reliance on the emergency services alone will not be sufficient to comply with the regulations.

If the fire and rescue service forms a part of the rescue plan, they must be given a warning that a confined space entry is to be made. This will give them the opportunity to assess the risks to their own staff and identify any equipment they might need.

8.6.1 Supervision

The level of supervision will be determined by the findings of the risk assessment. In most cases the level of risk will require the presence of a competent person to supervise the activities and remain present from start to finish. Supervisors should be given responsibility to ensure that the necessary precautions are taken to check safety at each stage.

8.6.2 Isolation

The confined space must be isolated from all possible external sources of danger to persons entering it. It is good practice to use **positive isolations** for confined space entries – this means complete separation of the plant or equipment to be worked on from other parts of the system. If complete isolation is not possible, an approved system must be in place to ensure an alternative isolation is effective and remains so.

A full permit to work system should be used to record the location and types of isolation, and the hazards being guarded against. Electrical isolation must never rely on a switch or fuse. The switch gear or fuse holder must be locked off and a warning notice applied *(for further information refer to 8.6.15 Permits).*

Mechanical isolation of pipework should not rely on a single valve or on a non-return valve; these may leak and create a hazard. Whenever possible, a section of pipe should be removed or a blank or spade should be put into a flange between the valve and the confined space and a warning notice displayed.

Isolation from mechanical and electrical equipment will often include locking off the switch and formally securing the key in accordance with a permit to work, until it is no longer necessary to control access. Lock-and-tag systems can be useful here, where each operator has their own lock and key giving self-assurance of the inactivated mechanism or system. Check there is no stored energy of any kind left in the system that could activate the equipment inadvertently.

Paddles, stirrers or agitators, whether electrically or mechanically operated, should be physically disconnected by the removal of an operating arm, and a warning notice displayed.

8.6.3 Cleaning

There is a variety of methods of cleaning the inside of confined spaces to remove hazardous solids, liquids or gases. Cold water washing, hot water washing and steaming will remove many contaminants, while solvents or neutralising agents may be necessary for others. If hot water or steam is used, with or without a solvent, care must be taken to ensure that adequate ventilation exists for steam pressure and that condensation does not build up to unacceptable levels.

If steam is used or water is boiled in a confined space, account must be taken of the vacuum that can be created on cooling. When steam or solvents are used, these may create a toxic, suffocating or flammable hazard. Even though a space has been well cleaned, it must not be entered until the atmosphere has been tested and is confirmed to be safe. Great care must be taken if encountering any sludge or heavy deposits that may release toxic gases if disturbed.

8.6.4 Purging and ventilation

Air purging and ventilation can be carried out by removing covers, opening inspection doors, or similar, and allowing ordinary air circulation, or by the introduction of compressed air via an air line. However, higher rates of air exchange can be achieved by the use of air movers, induction fans or extractor fans.

It is especially important that when an inert gas (such as nitrogen) has been used to purge or render inert a flammable atmosphere, the inert gas itself is properly purged with air. When air purging is taking place, the flow of air should be of a sufficient volume and velocity to ensure that no pockets or layers of gas remain undisturbed. This also needs to be constantly monitored.

8.6.5 Atmospheric monitoring

Depending on the circumstances, as a result of the assessment made under the COSHH Regulations, or the risk assessment made under the Management of Health and Safety at Work Regulations, continuous atmospheric monitoring may well be necessary when any work is to be done that would expose employees to any substance hazardous to health.

Before an entry is made into such a confined space, tests must be carried out to establish the levels of oxygen, toxic gas or flammable gas in the atmosphere.

The external atmosphere around the opening should be monitored first and, if the results are satisfactory, internal monitoring should be carried out by extending or lowering a gas monitor into the confined space before it is occupied. This may be necessary at different levels or points, depending on the size, height or depth of the space.

Atmospheric testing prior to entry

If entry into the confined space is necessary to carry out the tests, breathing apparatus or other RPE must be worn.

Suitably trained and competent personnel may use simple, reliable instruments to measure oxygen and flammable gas levels.

The accuracy of the instruments must be assured by periodic calibration.

A satisfactory oxygen content must not in itself be relied on to indicate safety since flammable, explosive or toxic gas may exist alongside oxygen and need only be present in minute quantities to create a serious hazard.

The tests applied should take account of what the space is known to have contained, including any inert gas used to purge a flammable atmosphere, which may itself produce toxic hazards or the risk of asphyxiation. Account must also be taken of hazards arising from other sources (such as materials used for cleaning). Methane, hydrogen sulphide and carbon dioxide can all evolve naturally due to the decomposition of organic matter or, in some cases, by the effect of rainwater percolating through certain types of ground. It is necessary to test the atmosphere of a confined space at both high and low level as well as in any corners where pockets of gas may exist.

Instances have occurred of carbon dioxide displacing oxygen at lower levels while a normal oxygen level continues to exist at higher levels of the same confined space.

The sense of smell must never be relied upon to detect gases. Some are odourless, and hydrogen sulphide, in particular, can paralyse the sense of smell to such an extent that even fatally high concentrations of the gas cannot be detected. The sense of smell varies from person to person and deteriorates with age.

8.6.6 Monitoring and testing equipment

Providing that the specific contaminant is known, tests can be carried out by competent persons using the individual detector tubes available for the detection of specific toxic and asphyxiant, flammable or explosive fumes or gases.

The tubes have a wide range of tolerance, so readings close to an occupational exposure limit should not be acted upon until the test has been repeated and the manufacturer's standard of accuracy applied.

A wide range of portable gas detection equipment is available for flammable and toxic gases; some are specific to one gas (for example, hydrogen sulphide), while others can sample a range of different gases. Such instruments need to be properly calibrated.

Close-up of a monitor

A monitor with individual detector tubes

8.6.6.1 Continuous monitoring

The initial monitoring and testing must establish that the confined space is safe to enter. Monitoring should then be carried out at intervals to ensure the continued safety of personnel. Tests should be repeated after any breaks (such as lunch or overnight) or after the time limit set out in a permit to work has expired.

It may be necessary to issue individual monitors to people working in a confined space to give them an instant warning of low oxygen, or toxic or flammable gas hazards.

8.6.6.2 Competence of monitors

All atmospheric monitoring must be carried out by persons who are trained and competent to use the instruments and interpret the results. They must have sufficient practical and theoretical knowledge to enable them to make a valid judgement based on the results. They must be fully aware of their responsibilities in permitting entry into a confined space.

8.6.7 Selection of personnel

Care is required in selecting the right people to work in confined spaces, since conditions can be difficult. They must be physically fit, agile and, most importantly, not be claustrophobic (fear of confined spaces). People who suffer from asthma, bronchitis or other respiratory conditions, or whose immune system has been suppressed, must be assessed by a medical practitioner as to their suitability to work in confined spaces.

Other health conditions that might indicate that a person is not suitable for working in a confined space, or that further checks need to be made before it is allowed, are listed below.

- High blood pressure.
- Partial or complete deafness.
- Lack of mobility through joint problems.
- Diabetes.
- Stress, depression or other mental ill health.
- Defective eyesight (which is not corrected by wearing glasses).
- Sensitivity of the skin to some substances.
- Taking some types of medication.

Stamina is also required. The wearing of any form of respiratory protection tends to lead to an increase in respiration and a higher use of energy; the amount of work that can be done in confined spaces is less than that expected under normal conditions.

When respiratory protection is to be used, it should be remembered that facial hair and spectacles often prevent a respirator from fitting properly and thus achieving the assumed degree of protection. Face-fit testing must be carried out to ensure that the chosen mask fits the wearer. As people come in all sorts of shapes and sizes, it is unlikely that one particular type or size of RPE face-piece will fit everyone.

Fit testing of individuals will ensure that the equipment selected is suitable for the wearer. This can be achieved quantitatively using a 'portacount' for half or full face masks or qualitatively using bitter/sweet solutions for half or disposable masks.

www **For more detail information on fit testing RPE, go to the HSE website to read or download their publication *Guidance on respiratory protective equipment (RPE) fit testing* INDG479.**

8.6.8 Communications

Adequate and effective communications must exist between those inside and those outside of the confined space, so that, in the event of an incident, a warning can be given and the space safely evacuated or those inside safely rescued. The system needs to be fail-safe, ensuring that, if a reply is not received or a scheduled call not made, the procedure for rescue starts immediately.

When a confined space is relatively small, so that the person entering it cannot move far from the entry point and there are no other factors that could hinder effective communication, the method of communication may be relatively simple (such as a prearranged system of tugs on the safety rope, which must be fully understood by all involved). However, if the nature of the confined space, the job to be carried out and other factors necessitate the person travelling some distance from the entry point, a more elaborate communication system might be required.

The following factors could hinder effective communication and may need to be considered in the risk assessment.

- Level of noise inside or outside of the confined space, which may or may not be associated with the confined space work.
- Physical nature of the confined space or the presence of substances that could reduce visibility.
- Distance between the point of entry and the place of work.
- Presence of workers with little or no understanding of English.

Depending upon the findings of the risk assessment, prior communication with the emergency services regarding the location and nature of the work might be considered necessary. This will allow the emergency services to comment on the rescue plan and also prepare for any eventuality. All types of respiratory protection affect verbal communication to some degree and, whatever method of communication is chosen, it should be tested and proved outside the confined space before entry is made.

8.6.9 Work equipment

Due to the potential for a flammable or explosive atmosphere in confined spaces, selecting tools and other work equipment with which the work can be carried out safely is essential.

If there is any possibility of flammable gas existing in a confined space, even below the lower explosive limit (LEL), all tools must be of a non-sparking material (for example, non-ferrous materials such as brass, bronze or copper-aluminum alloys – these softer materials greatly reduce the risk of producing ignitable sparks) and all lighting and electrical equipment should be extra low voltage (less than 25V) and must carry British Approvals Service for Electrical Equipment in Flammable Atmospheres (BASEEFA) approval.

Smoking and naked lights must be strictly prohibited, and care must be taken to avoid the generation of static electricity with the consequent risk of sparks.

8.6.10 Fire safety

Hot works must not be carried out in a confined space unless atmospheric testing has confirmed that flammable or explosive gases are not present and the findings of a risk assessment show that it is safe to do so.

Where there is still a residual risk of fire, appropriate fire extinguishers may need to be kept in the confined space at the entry point. Operatives must be suitably trained in the selection and use of appropriate fire extinguishers.

Where hot works are being carried out inside a confined space, the operative carrying out the work must also have a suitable and serviceable fire extinguisher at the place of work. In the event of a fire, the local fire service should be called in case the fire cannot be contained or extinguished.

8.6.11 First aid

Appropriate first-aid equipment and suitably trained first aiders should be provided and available for emergencies and to provide first aid until professional medical help arrives. First-aid training should cover incidents that may occur as a result of the hazards present in the confined space.

8.6.12 Rescue

The arrangements for the rescue of persons in the event of an emergency, both in terms of trained persons and equipment, need to be carefully planned and practised. The rescue plan must be able to be implemented swiftly in an emergency and understood by all involved. The arrangements must be in place and all equipment checked before any person enters or works in a confined space.

In some circumstances (for example, where there are prolonged operations in confined spaces and the risks justify it) there may be advantage in prior notification to the local emergency services (for example, local fire or ambulance service) before the work is undertaken.

However, reliance on the emergency services alone will not be sufficient to comply with the regulations. Employers must put in place adequate emergency arrangements before the work starts. Proper and effective rescue training is quite hard and arduous, and is not to be undertaken lightly. Persons selected for such training need to be physically fit and able to adapt to situations as they arise during a rescue.

Injury and collapse

If a person is **injured** in a confined space that has been certified safe to enter without respiratory protection, an entry can be made to rescue and remove them straight away.

If a person **collapses** in a confined space and the cause is not known, irrespective of whether or not the confined space was certified fit for entry without respiratory protection, no-one must enter unless they are wearing breathing apparatus. The collapse may have been due to deterioration in the atmosphere within the confined space. The first duty of any rescuer is to ensure that they do not become a casualty themselves.

Each year, would-be rescuers who are insufficiently trained or equipped die by going into confined spaces where a person has collapsed.

This point is illustrated in the following two case studies.

Open-topped inspection shaft fatalities

At Carsington Reservoir, four men, all aged between 20 and 30 and physically fit, died in an open-topped inspection shaft. Naturally evolved carbon dioxide had displaced the oxygen. No tests were made before entry. The first man down collapsed and the three other men climbed down to their deaths in futile attempts to rescue him.

An HM Inspector of Factories working in the Public Utilities National Industry Group of the HSE wrote about this major accident and stressed: 'It should be manifestly obvious that confined spaces working is one activity where shortcuts in safety cannot be permitted. The deaths of these four men will be more tragic if the lessons of this incident are not fully learnt'.

Fatalities in a sewer

When an engineer collapsed in a sewer, a rescuer entered without breathing apparatus and was overcome. A second person made a similarly vain attempt to reach the victims. When the rescue team from the fire and rescue service arrived they had to remove the bodies of the two would-be rescuers before they could get to the engineer. By then it was too late and he had died. Had everyone waited for the fire and rescue service, the engineer might have lived and the rescuers would not have needlessly died.

8.6.13 Rescue equipment

Every person entering a confined space wearing breathing apparatus must also wear a rescue safety harness. The harness must be attached to a lifeline, attended by a person outside the confined space.

The harness must be one that is suitable for confined space rescue, in that it must enable an unconscious person to remain in an upright position whilst being hoisted.

An unsuitable harness will allow the unconscious person to bend at the waist, making recovery through a narrow opening difficult or impossible.

This equipment forms part of a safe system of work for any entry into a confined space. Properly used, it may enable a rescue to be carried out successfully without the need for a rescuer to enter the confined space.

Rescue equipment must include some means of lifting or pulling a person up from a confined space, since it is virtually impossible for the average person to achieve this solely by muscular effort. There are a variety of tripods, winches, blocks and tackles which, when used in conjunction with a safety harness, enable a person to be lifted quickly and safely out of a confined space.

If any lifting equipment or accessories are to be used, then the equipment and accessories will be subject to the Lifting Operations and Lifting Equipment Regulations 1998 (LOLER). This would mean testing and inspection in accordance with the schedule drawn up by the competent person.

In practice, harnesses, lines and accessories (such as carabiners) should be subjected to a formal, thorough examination by a competent person every six months and be checked by the user weekly and before each use. Tripods, hoists and other lifting devices need to be load tested every six months, as part of the thorough examination, in the same way that a scissor lift used for lifting people would.

Dependent on circumstances, rescue equipment may have to include first-aid equipment, oxygen or resuscitation packs and rescue breathing apparatus. A secure line of communication to the emergency services may also be required.

8.6.14 Respiratory protective equipment

RPE is provided in addition to engineering controls and safe systems of work. If it is identified in the risk assessment that RPE is required, the RPE must be selected by a competent person, be CE marked and be suitable for the type of hazard against which it is to protect the wearer.

A wide range of RPE is available from various manufacturers and PPE suppliers. There are two main types of RPE for use in confined spaces work, which are outlined below.

8.6.14.1 Respirators – Purifying the air breathed

The air inhaled is drawn through a filter or medium that removes the harmful substance or pollutant. The nature of the filtering agent depends on the type of pollutant to be dealt with. These types are commonly called *respirators*.

The simplest form of respirator is the dust mask, a preformed cup made of filtering material that fits over the nose and mouth to filter out nuisance dust. These masks give no protection against harmful or toxic gases or fumes and the protection factor of the mask may not offer adequate protection against the level of airborne dust that can be experienced in a confined space.

More complex types have filter cartridges that may be suitable for various types of dust or fume, or are specific to a particular substance.

8.6.14.2 Breathing apparatus – Supplying clean air

The air can be supplied straight through an air line via a pump or compressor or, alternatively, the person may carry compressed air in cylinders. These types are known as *breathing apparatus*.

An alternative type of breathing equipment is the escape breathing apparatus or self-rescue set. These can consist of a small, compressed air bottle, the necessary hoses and valves and a facepiece. Self-rescue sets can be carried by operatives who enter confined spaces where the air is initially safe to breathe.

Should the air quality deteriorate, the facepiece is placed over the nose and mouth and the air valve opened. The air bottle supplies fresh air to the operative whilst an escape from the confined space is made. The air bottle of a self-rescue set has a duration of typically 15-20 minutes.

Other devices can supply oxygen via a chemical reaction. Some types can provide up to 60 minutes of rescue oxygen.

Rescue training with dead weight rescue dummy and rescue sets

Care must be taken to select the correct type of protection for the conditions. Respirators (as opposed to breathing apparatus) do not protect against oxygen-deficient atmospheres and should not be used in any atmosphere dangerous to life.

RPE should not be used unless all other methods of control or protection have been examined and it is established that the use of RPE is the only reasonably practicable solution.

The HSE publication *Respiratory protective equipment at work – A practical guide* (HSG53) sets out the nominal protection factor for each type of respirator and describes their limitations; it should be carefully consulted in cases of doubt.

Respirators can only be used for protection against the gases or dusts for which they are specifically intended. It is important to note that all forms of RPE will have a limited period of usage before becoming saturated with the contaminant. They may also have a limited shelf life, indicated by a use-by date.

8.6.15 Permits

8.6.15.1 Permit to work

A permit to work ensures a formal check is undertaken to make sure all the elements of a safe system of work are in place before people are allowed to enter or work in the confined space.

It is also a means of communication between site management, supervisors and those carrying out the hazardous work.

Entries into a confined space can be made under a permit to work, whereby a competent person must be satisfied that all necessary precautions have been taken and provisions made to secure the safety of those entering the confined space, before signing the permit to work. The signed permit thus gives an assurance that work may safely take place.

Permits should only be issued by named, authorised persons, who must sign them. Such persons must be competent, have authority and possess sufficient practical and theoretical knowledge and actual experience of working conditions to enable them to judge whether everything necessary has been done to ensure the safety of personnel.

It is quite common for several authorised persons to sign a permit to work, each certifying that they have taken the necessary actions with regard to their own area of responsibility (for example, electrical isolation and atmospheric testing).

Where a permit to work system involves the use of padlocks and keys (for example, for locking-off electrical isolators or other sources of energy) the keys must stay with authorised persons until such time as the permit is returned for cancellation, either when the task is completed or when any time limits stated on the permit have been reached (such as at the end of a shift).

Essential features of a permit to work are listed below.

- Clear identification of who may authorise particular jobs (and any limits to their authority) and who is responsible for specifying the necessary precautions (for example, isolation, air testing and emergency arrangements).
- Making sure that contractors engaged to carry out work are included.
- Training and instruction in the issue of permits.
- Monitoring and auditing to make sure that the permit to work system works as intended.

8.6.15.2 Permit to enter

Depending upon the nature of the confined space and the inherent risks of carrying out the work, some companies may choose to run a separate permit to enter system. An example of when such a system might be used is where all preparatory work is carried out to meet the requirements of the permit to work and then the permit to enter is issued when final pre-entry checks of the atmosphere have been carried out.

Such a system would cover situations where a single permit to work covers the duration of the whole job, but successive shifts of workers are each authorised to enter the confined space under a newly raised permit to enter.

8.6.15.3 Access and egress

A safe way in and out of the confined space should be provided and, wherever possible, allow quick, unobstructed and ready access (such as a fixed, vertical ladder inside an underground chamber that terminates just below the entry/exit point at ground level).

The means of escape must be suitable for use by the number of individuals who enter the confined space so that, ideally, they can quickly escape in an emergency. However, it must be accepted that in many cases the entry/exit point will be of a restricted size that will not necessarily allow an easy escape route in an emergency, particularly if the person who is escaping is wearing a compressed air cylinder. The means of achieving a prompt escape or rescue must be considered in the risk assessment.

Suitable means to prevent access (for example, a locked hatch) should also be in place when there is no need for access to the confined space. There should be a safety sign that is clear and conspicuous to prohibit unauthorised entry alongside openings that allow for safe access.

8.6.16 Conclusion

For work to be done safely in a confined space, great care has to be taken over the detail of each step of the procedure. Common causes of accidents are shown below.

- Failing to identify work areas as confined spaces.
- Failing to identify activities or processes that can affect the atmosphere of a safe area, resulting in it becoming an unplanned confined space.
- Inadequate risk assessments.
- Poorly trained and equipped workers.
- Failing to put in place adequate emergency arrangements before work starts.
- Failing to set up a safe system of work, including continuous air monitoring, based around a permit to work system.
- Failing to carry out an initial check of air quality.
- Failing to follow an established safe system of work.
- Deviations in conditions rendering the risk assessment or safe system invalid.
- Incorrectly using RPE.
- Using the wrong type of RPE.
- Failing to use safety harnesses and lifelines.
- Ill-conceived and badly executed rescue attempts.

All such accidents are avoidable. If an accident does occur, it demonstrates that a breakdown has occurred in the supposed safe system of work.

8.7 Legislative requirements

In addition to the **Confined Spaces Regulations** and the **Management of Health and Safety at Work Regulations** the following legislation may also apply.

8.7.1 Construction (Design and Management) Regulations 2015

These regulations place a legal duty on designers, when preparing designs, to design out risk, so far as it is reasonably practicable.

In the context of this chapter, designers should carry out their design work so that no-one has to enter a confined space during construction work, maintenance or cleaning of the structure or during its demolition.

The regulations place a duty on the designer, when preparing their designs, to consider the users of any finished structure that will be a workplace.

Also, within the context of this chapter, these regulations place legal duties on contractors and principal contractors, where applicable, with regard to the following.

- Safe places of work.
- Cofferdams and caissons.
- Excavations.
- Prevention of drowning.
- Prevention of risk from fire, explosion, flooding and asphyxiation.
- Emergency procedures.
- Fresh air.

For further information refer to Chapter A03 Construction (Design and Management) Regulations 2015.

08

8.7.2 Provision and Use of Work Equipment Regulations 1998

These regulations require that an employer only supplies work equipment that is correct and suitable for the task and work environment. PUWER require that the equipment is inspected, maintained and kept in good working order. Where the use of the equipment involves a specific risk to the health and safety of employees, the use of the equipment must be restricted to specified trained and authorised workers.

8.7.3 Personal Protective Equipment at Work (Amendment) Regulations 2022

These regulations require that, where a risk has been identified by a risk assessment and it cannot be adequately controlled by other means, which are equally or more effective, then the employer must provide and ensure that suitable PPE is used by employees.

In essence, PPE may only be used as a last resort after all other means of eliminating or controlling the risk have been considered and are found to be not reasonably practicable to implement. In practice, however, unless it is possible to carry out the work without entry into the confined space, the wearing of PPE will usually be necessary.

In deciding which type to issue, the employer must take into account the risk that the PPE is being used to protect against, and ensure that the PPE will fit the wearer and allow them to work safely. Where the use of RPE is necessary, face-fit testing to establish the suitability of the RPE for the wearer is required. If more than one item of PPE is being used, the employer must make sure that individual items of PPE are compatible and suitable for the task that is to be undertaken.

Whenever PPE is to be issued, the employer must ensure that employees have been given adequate and appropriate information, instruction and training to enable them to understand the risks being protected against, the purpose of the PPE and the manner in which it is to be used.

Whilst the employer must ensure that PPE is supplied and used, the employee has a duty to properly use the equipment provided, follow the information, instruction and training that they have been given, and know the procedures for reporting losses or defects to their employer. In the context of this chapter, in addition to the more commonly used PPE, confined space working will often require the use of appropriate RPE and rescue equipment (such as a safety harness and line).

For further information on the Personal Protective Equipment at Work Regulations refer to Chapter B06 Personal protective equipment.

8.7.4 Lifting Operations and Lifting Equipment Regulations 1998

Access to and egress from many confined spaces is made by lowering or raising a person vertically through the entry or exit point, including during practice or actual rescues. In these circumstances:

- safety harnesses and rescue lines must be regarded as lifting accessories
- the tripod hoist or other type of winch must be regarded as lifting equipment used for lifting persons and, as such, requires thorough examination on a six monthly basis under LOLER.

However, if the rescue involves a sideways or inclined drag (for example, along a duct) then there would be no lifting and LOLER would not apply, although the Provision and Use of Work Equipment Regulations would.

For further information on LOLER refer to Chapter C07 Lifting operations.

8.7.5 Other legislation

The following legislation can also have an impact upon how work in confined spaces is planned, organised and carried out.

- Health and Safety (First-Aid) Regulations 1981 *(refer to Chapter B05 First aid).*
- Control of Substances Hazardous to Health Regulations *(refer to Chapter B07 Control of substances hazardous to health).*
- Work at Height Regulations 2005 *(refer to Chapter D01 Work at Height Regulations 2005).*
- Dangerous Substances and Explosive Atmospheres Regulations 2002 *(refer to Chapter D09 Dangerous substances).*
- Regulatory Reform (Fire Safety) Order 2005 (England and Wales only) *(refer to Chapter C02 Fire prevention and control).*

CONTENTS

Dangerous substances

09

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.

Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

Dangerous substances are widely used on construction sites. The main hazards are fire and explosion, and everything possible must be done to lessen the risks.

Safety is divided into three main areas: the storage of substances, the safe handling and transport of substances and the uses to which substances are put.

9.1 Introduction

This chapter does not cover the storage of explosives, which may be used in the demolition and civil engineering sectors. There are specific requirements under the **Explosives Regulations 2014**. The regulations are supported by guidance which is available from the Health and Safety Executive (HSE).

In most cases, the HSE enforces fire safety legislation on construction sites, under the Construction (Design and Management) Regulations 2015 (CDM) and the Management of Health and Safety at Work Regulations 1999, although they also have enforcement powers under the Regulatory Reform (Fire Safety) Order 2005 (FSO) or in Scotland under the Fire (Scotland) Act 2005. In Scotland requirements on general fire safety are covered in Part 3 of the Fire (Scotland) Act. The Fire Safety Regulations (Northern Ireland) apply in Northern Ireland.

Local Authority environmental health officers are likely to have a similar role where the work is carried out on a construction project in a premises that they enforce. The local fire and rescue service (FRS) has responsibilities for the enforcement of fire safety in some circumstances.

Dangerous substances and explosive atmospheres

Dangerous substances. Any substance or mixture of substances that can put people's health or safety at risk from fire and explosion.

Explosive atmospheres. Any substance or mixture of air and one or more hazardous substances in the form of a gas, vapour, mist or dust, which will explode after ignition has occurred.

9.2 Important points

- A risk assessment must be carried out before dangerous substances are stored, transported or used. In some cases, a method statement and/or a permit to work will also be required.
- Important pieces of legislation relating to this chapter are the:
  - Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR)
  - CDM
  - FSO (England and Wales only)
  - Fire (Scotland) Act
  - Fire Safety (Employees' Capabilities) (England) Regulations 2010
  - Fire Safety Regulations (Northern Ireland) 2010
  - Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations 2009 (CDG).
- DSEAR covers the flammable or explosive properties of dangerous substances used in the workplace.
- There may also be health issues that are covered by the Control of Substances Hazardous to Health Regulations 2002 (COSHH Regulations).
- Fire safety legislation will be enforced by the HSE or the local FRS, generally depending upon the location where dangerous substances are stored and used.
- People who use dangerous substances must be fully aware of their hazardous properties, adopt methods of controlling the risks and be trained in the selection and use of portable fire extinguishers.
- Electrical apparatus and naked flames should not be used near flammable or explosive dangerous substances, particularly if they are being sprayed.

Uncontrolled use or storage of explosive materials can have fatal consequences

- Good ventilation is essential wherever flammable or explosive dangerous substances are used or stored.
- Non-smoking policies and waste disposal policies must be established and diligently monitored.
- Liquefied petroleum gas (LPG) cylinders are fitted with a spring loaded safety relief valve that will lift and vent gas into the atmosphere if the internal gas (head) pressure valve reaches 26 bar.

Start by identifying dangerous substances on site and quantities. Use the material safety data sheets and labels to identify these hazards. Also consider processes that may generate dangerous forms of substances.

9.3 Dangerous Substances and Explosive Atmospheres Regulations 2002

9.3.1 Principles of the regulations

DSEAR require employers to control the risks to safety from fire and explosions.

The regulations apply at all places of work where a dangerous substance is present (or is liable to be present) or if the dangerous substance could be a risk to the safety of people as a result of fires, explosions or similar energetic events.

Dangerous substances are:

- a substance or mixture of substances that is classified as explosive, oxidising, extremely flammable, highly flammable or flammable
- any dust, whether in the form of solid particles or fibrous materials, which can form an explosive mixture in air.

In the construction industry, many dangerous substances are used, or created by work. Some examples are shown below.

- The storage and use of solvents, adhesives and paints.
- The storage and use of flammable gases (such as oxygen and acetylene) during cutting and welding.
- The storage and use of LPG for work processes, heating or cooking.
- The creation of large quantities of airborne dust (for example, as a result of wood-machining or sanding) and the handling and storage of bulk waste dust.
- The storage and decanting of vehicle fuels and lubricants.
- The storage and handling of liquid flammable wastes (such as fuel oils).
- Many tasks that involve hot works (such as the hot-cutting of tanks and drums that have contained flammable materials).

www **For further guidance on DSEAR visit the HSE website.**

9.3.2 Employers' duties under DSEAR

DSEAR place duties on employers (and the self-employed, who are considered employers for the purposes of the regulations) to assess and eliminate or reduce risks from dangerous substances.

Complying with DSEAR involves the responsibilities shown below.

9.3.2.1 Assessing risks

Where a dangerous substance is, or is liable to be, present at the workplace, employers must assess the fire and explosion risks that may be caused by dangerous substances. This should be an identification and careful examination and as a minimum should:

- identify and determine the hazardous properties of the dangerous substance(s)
- identify those different groups of workers and people who may be harmed, and the likelihood and severity of the consequences
- consider any employees who may be at increased risk because of lack of awareness, such as inexperienced trainees and those under 18
- consider others, including workers of another employer in the workplace or nearby, members of the public and other visitors, both on and off site
- be satisfied that, where a 'model' risk assessment is being used from somewhere else that uses similar processes, in each case, the model:
  - reflects the core hazards
  - is adapted to the details of the particular situation
  - is appropriate to the type of work.

The purpose is to help employers to decide what they need to do to eliminate or reduce the risks from dangerous substances. If there is no risk to safety from fires and explosions, or the risk is low, no further action is needed. If there are risks then employers must consider what else needs to be done to comply fully with the requirements of DSEAR. If an employer has five or more employees, the employer must record the significant findings of the risk assessment. This is the same risk assessment required under the Management of Health and Safety at Work Regulations. There is no requirement to carry out two risk assessments, as long as the requirements of both sets of regulations are met.

The assessment (including the recording of significant findings) enables employers to demonstrate to themselves, and to others who may have an interest (such as HSE inspectors, principal contractors and employees' representatives), that they have followed a structured and thorough approach in considering the risks to the safety of employees and the control measures that are needed. The persons carrying out the assessment should have the skills, knowledge, experience and training (competence) to do so.

9.3.2.2 Preventing or controlling risks

Employers must put control measures in place to eliminate risks from dangerous substances, or reduce them as far as is reasonably practicable. Where it is not possible to eliminate the risk completely employers must take measures to control risks and reduce the severity of the effects of fire or explosion.

The best solution is to eliminate the risk by replacing the dangerous substance with a less dangerous substance, or using a different work process. In DSEAR, this is called substitution.

In practice, this may be difficult to achieve. However, it may be possible to reduce the risk by using a less dangerous substance (for example, by replacing a low flashpoint liquid with a high flashpoint liquid). In other situations, it may not be possible to replace the dangerous substance. For example, it would not be practical to replace petrol with another substance at a filling station.

9.3.2.3 Control measures

Where the risk cannot be eliminated, DSEAR require control measures to be applied in the following priority order.

1.	Reduce the quantity of dangerous substances to a minimum.
2.	Avoid or minimise releases of dangerous substances.
3.	Control releases of dangerous substances at source.
4.	Prevent the formation of a dangerous atmosphere, including the application of appropriate ventilation.
5.	Collect, contain and remove any releases to a place that is safe or is otherwise rendered safe (for example, through ventilation).
6.	Avoid ignition sources.
7.	Avoid adverse conditions (for example, exceeding the limits of temperature or control settings) that could lead to danger.
8.	Keep incompatible dangerous substances apart.

These control measures should be proportionate to the degree of risk as highlighted in the risk assessment and be appropriate to the nature of the activity or operation.

Worker suffers severe burns during equipment refuelling

A company specialising in spray foam insulation services has been fined after an inexperienced worker suffered severe burns during the refuelling of petrol powered equipment.

Southwark Crown Court heard how workers were spraying insulation into a ceiling cavity of a retail outlet. The foam spraying equipment was installed in a van parked outside of the premises. The worker entered the van to refuel the equipment from a jerry can container fixed with straps, within the compartment containing a petrol powered compressor/generator. When the jerry can was opened, petrol sprayed over the worker and the vapour ignited immediately, thereby covering him in flames. He was in a coma for three months and spent over a year in hospital.

HSE investigators found that the company had failed to ensure that risk from petrol was either eliminated or reduced so far as is reasonably practicable. Reasonably practicable actions, which the HSE argued could have been taken, included using diesel powered spray foam equipment or reducing the frequency of refuelling by installing a larger fuel tank or tanks. Refuelling could then have been reduced to once a day and could have taken place at the beginning of the day when the equipment was cool and not in operation. The potential for petrol spillage could have been reduced by storing it away from sources of heat and confining it to smaller containers, or by using a non-spilling fuel delivery nozzle.

The insulation company from Birmingham pleaded guilty to breaching Regulation 6 of the Dangerous Substances and Explosive Atmospheres Regulations 2002. The company was fined £40,000 and ordered to pay costs of £11,779.

Speaking after the hearing an HSE inspector said: 'This was the worker's second day on the job. He suffered horrific injuries due to the company's failure to adequately consider the risks from refuelling and implementing safer alternatives to the system of work requiring refuelling petrol powered equipment every two hours'.

(Source: HSE.)

9.3.2.4 Mitigation

In addition to control measures, DSEAR require employers to put mitigation measures in place. These measures should be consistent with the risk assessment and appropriate to the nature of the activity or operation, and include the following.

- Reducing the number of employees exposed to the risk.
- Providing explosion-resistant plant.
- Providing explosion suppression equipment.
- Providing explosion pressure relief arrangements.
- Taking measures to control or minimise the spread of fires or explosions.
- Providing suitable personal protective equipment (PPE).

9.3.2.5 Preparing emergency plans and procedures

Arrangements must be made to deal with emergencies. These plans and procedures should cover escape routes, safety drills, first aid and suitable communication and warning systems, and should be in proportion to the risks. Where possible visual or audible warnings should be initiated so employees can be withdrawn before explosion conditions are reached. If an emergency occurs only workers essential to carrying out repairs must be allowed into the area and they must be provided with the appropriate PPE and equipment and plant to allow them to carry out this work safely.

The information in the emergency plans and procedures must be made available to the emergency services to allow them to develop their own plans if necessary.

9.3.2.6 Providing information, instruction and training for employees

Employees must be provided, in an appropriate manner, with relevant information, instruction and training, which will include the following.

- The dangerous substances present in the workplace and the risks they present, including access to any relevant safety data sheets and information on any other legislation that applies to the dangerous substance.
- The findings of the risk assessment and the control measures put in place as a result, including their purpose and how to follow and use them.
- Emergency procedures.

Information, instruction and training need only be provided to other people (non-employees) where it is required to ensure their safety. It should be in proportion to the level and type of risk.

The information, instruction and training provided should be appropriate to the level of understanding and experience of the employees. It should be provided in a form which takes account of any language difficulties or disabilities. Information can be provided in whatever form is most suitable in the circumstances, as long as it can be understood by everyone.

In practical terms, the information must include the following.

- How and where the dangerous substance is to be used in the specific site activities in addition to the general information in the safety data sheet.
- The precautions and actions required. The information for employees should include the control and mitigation measures adopted, including methods of work, the reasons behind them, and how to use them properly.
- Training and instruction, which should include the reasoning (theory) behind the practice. Training in the use and application of control measures and equipment should be carried out, taking into account recommendations and instructions supplied by the manufacturer.
- Any procedures for dealing with accidents, emergencies and incidents. This ranges from smaller unplanned incidents (including dealing with faults and clearing blockages) to larger emergencies and should prepare staff for how to react if and when foreseeable events happen.
- Any further relevant information resulting from a review of the risk assessment, why it has been done and how any changes will affect the way employees do the work in the future.

Employers may need to make special arrangements for employees with little or no understanding of English, or those who cannot read English. This could include providing translation, using interpreters or replacing written notes with clearly understood symbols or diagrams.

Employers also need to take account of the needs of people other than employees who may be present on site, such as contractors, visitors or members of the public. While it may not always be practical to provide formal training in these circumstances, employers should consider what other information or instruction may be needed at site inductions to reduce risks.

For example, this could include pictorial signs for infrequent visitors to the site or for those whose first language is not English (which might be the case for delivery drivers and other staff or visitors), notices explaining hazards (warning notices, no smoking signs and so on) and copies of emergency and evacuation procedures.

The contents of pipes and containers must be identifiable to alert employees and others to the presence of dangerous substances, so that they can take the necessary precautions. It can also avoid incorrect mixing of contents.

9.3.2.7 Places where explosive atmospheres may occur

DSEAR places additional duties on employers where potentially explosive atmospheres may occur in the workplace. In relation to construction site work, this could include bottled gas, petrol or fuel storage areas. These duties include the following.

- Identifying and classifying (zoning) areas where potentially explosive atmospheres may occur.
- Avoiding ignition sources in zoned areas, in particular those from electrical and mechanical equipment.
- Where necessary, identifying the entrances to zoned areas by the display of signs.
- Providing appropriate anti-static clothing for employees.
- Before they come into operation, verifying the overall explosion-protection safety of areas where explosive atmospheres may occur.

Decisions on the zoning of areas and the appropriate actions to take must be made by someone who has been trained and is competent to do so.

Note: *The above summary of DSEAR is reproduced from the HSE website under licence from The Controller of Her Majesty's Stationery Office.*

Whilst the employer must, as far as possible, ensure that any PPE supplied must be worn, the employee in turn must ensure that they wear and use the equipment provided correctly and know the procedures for replacing and reporting loss or defects to the employer.

In the context of this chapter, the relevance of these regulations includes the prevention of:

- inhalation of fumes and vapour given off by dangerous substances
- skin contact with dangerous substances
- eye injuries resulting from splashes of dangerous substances.

For further information on the Personal Protective Equipment at Work Regulations refer to Chapter B06 Personal protective equipment.

9.4 Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations 2009, and the European agreement

The Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations (CDG) and the European agreement (ADR) are highly prescriptive and deal with the carriage of dangerous goods and their purpose to protect everyone, either those directly involved (such as consignors or carriers) or those who might be harmed (such as members of the public or the emergency services).

Note: *ADR comes from the French abbreviation **A**ccord Européen relatif au transport international des marchandises **D**angereuses par **R**oute.*

Dangerous goods are liquid or solid substances, and the articles containing them, that have been tested, assessed and given a classification. Dangerous goods are given different classes depending on their predominant hazard. Goods being carried by rail or road are dangerous if there is a risk of spillage leading to fire, explosion, chemical burn or environmental damage following an incident. The regulations place duties upon everyone involved in the carriage of dangerous goods to ensure they know what they have to do to minimise the risk of incidents and guarantee effective responses.

Many duty holders will need to appoint a dangerous goods safety adviser.

9.4.1 CDG

These regulations were substantially restructured in 2009, with direct referencing to ADR for the main duties, with amending regulations made in 2011. CDG now cross-references almost totally to ADR; it is ADR that contains the detailed requirements. CDG sets the legal framework in the UK; ADR has no provision for enforcement. CDG includes a number of exemptions and make substantial changes to ADR requirements for the domestic (within the UK) carriage of many explosives.

Note: *At the time of publication, the current CDG regulations are undergoing review and amendments by the UK Government, to meet the requirements of the UK's withdrawal from the EU.*

9.4.2 ADR structure

Each part of ADR is sub-divided into chapters, with each chapter further divided into sections and paragraphs.

Part number	Description
1.	Introduction – setting out high-level aims and duties, together with exemptions. This part includes the need for a dangerous goods safety adviser.
2.	Classification.
3.	The dangerous goods list (including special provisions and exemptions related to limited quantities).

Part number	Description
4.	Packing and tank provisions.
5.	Consignment procedures, including documentation and vehicle marking.
6.	Construction and testing of packages, intermediate bulk containers, large packages and tanks.
7.	Carriage – loading, unloading and handling.
8.	Vehicle crews, equipment, operation and documentation (including driver training).
9.	Construction and approval of vehicles.

For further information on the requirements of CDG and ADR visit the HSE website.

9.5 Control of Substances Hazardous to Health Regulations 2002

The COSHH Regulations only apply to dangerous explosive or flammable substances where they also possess other hazardous properties. This would be identified as part of the COSHH assessment carried out on the substance.

For further information refer to Chapter B07 Control of substances hazardous to health.

9.6 Storage of dangerous substances

On most construction sites, dangerous substances will be used at some time during the construction phase.

Depending upon the nature of the work it may be necessary to store bulk quantities of dangerous substances. Where site conditions allow this should be in an external, secure, purpose-built compound, or where this is not possible in a suitable, secure internal storeroom.

Alternatively, small quantities (for example, 200 ml containers upwards) will often be taken to the place of work by the person doing the job. Where small quantities of dangerous substances for daily use are required in the workplace, metal lockable bins may be used.

Safe and secure storage

Gas cylinders are capable of withstanding normal UK weather conditions, including temperatures between -20°C to +55°C. As such, they may not require protection from the weather when in storage. The floor should be constructed of concrete or other rigid, permanent, non-combustible, non-porous material. The floor should not be constructed of materials such as bricks, slabs, flagstones, asphalt or tarmacadam.

9.6.1 Storage in the open air

Where it is necessary to store dangerous substances in bulk, a store should be built to the following requirements.

- On a concrete sloping pad with a sump to catch any leaks or spills.
- With a low sill to create a bund with the capacity equal to 110% of the contents of the largest can or drum being stored.
- Surrounded by a 1.8 m high wire fence.
- So that it is protected against direct sunlight.
- At least 2 m away from nearby buildings or boundaries, except that, where the boundary of the store forms part of a solid wall, cans or drums may be stacked up against that wall up to 1 m from the top.

Cans or drums should be stored so that their contents can be easily identified and removed in the event of any leak or damage. They must also be stored on their sides and chocked to prevent movement. Stores or bins must be kept locked and only sufficient amounts for each day's requirements should be removed, as and when needed.

They may be marked with suitable signs, such as 'Flammable liquid' or 'Flammable gas'. Additionally, if an assessment under DSEAR shows that an explosive atmosphere may be present in a particular area, appropriate numbers of the illustrated sign must be displayed.

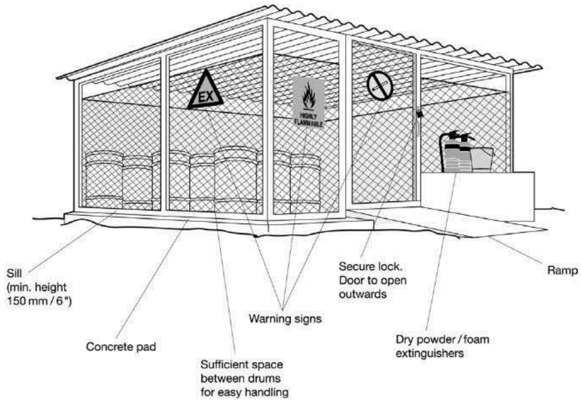
Examples of suitable warning signs

The sign comprises a yellow background and black graphics. Signs must conform to the Health and Safety (Safety Signs and Signals) Regulations 1996 and *Safety signs* (BS 5499). All signs purchased from reputable suppliers will be to these specifications.

Naked flames, smoking, matches or lighters must not be allowed in the area of the store, and proper prohibition signs must be clearly displayed, as well as other signs already indicated.

Any lighting within a store must be flameproof to the appropriate standard, and under no circumstances should electrical sockets be permitted.

Where there is a need for electrical apparatus (other than lighting) within a store, the supply must be permanently wired in using intrinsically safe equipment. Non-intrinsically or explosive safe-rated electronic devices (such as mobile phones, walkie talkies and torches) must not be allowed in the area of the store, and proper prohibition signs must be clearly displayed in addition to the other signs already indicated.



Example of a secure storage area

Other points to be noted.

- Stores should not be built below ground level because the vapours from spills and leaks will not be able to disperse (they are heavier than air).
- Stores must be located away from any emergency exit routes.
- Adequate cross-ventilation at both high and low level is necessary.
- The store should always be kept locked when unattended.
- A door sill of approximately 150 mm in height should be provided to catch any leaks or spills in order to retain the liquids inside the building.
- A sufficient quantity of absorbent material, to soak up any spilt liquids, and suitable containers for the collection and safe disposal of the contaminated absorbent, should be provided at the store.
- The use of mobile phones in or immediately adjacent to the store should be prohibited. Notices to this effect should be provided and prominently displayed.
- Any shelves or racking in the store should be of a non-ferrous metal or other non-combustible construction.

9.6.2 Storage inside a building

A separate building should be provided, if possible, solely for the storage of dangerous substances where security or protection from the weather is required. Ideally, it will be constructed from fire-resisting materials and it should be at least 2 m away from other buildings or boundaries.

A risk assessment should be carried out to determine whether the risks of storing dangerous substances in such a location are acceptable. If not, either additional control measures must be put in place or alternative arrangements made for storing the substances.

Where a separate building cannot be provided, and the store forms part of an existing structure, the surrounding walls and roof of the store must be fire-resisting and the door should be of the fire-resisting type and open outwards.

It is recommended that the maximum quantities that may be stored in cabinets and bins are no more than 50 litres for highly flammable liquids (and flammable liquids with a flashpoint below the maximum ambient temperature of the workroom/working area) and no more than 250 litres for other flammable liquids with a higher flashpoint of up to 55°C.

Other points to be considered.

- Stores should not be built below ground level, because the vapours from spillages and leaks will not be able to disperse.
- Adequate cross-ventilation at both high and low level is necessary.
- The store should always be kept locked when unattended.
- When not in use, containers of flammable liquids needed for current work should be kept closed and stored in suitable cabinets or bins of fire-resisting construction, which are designed to retain spills (110% volume of the largest vessel normally stored in it).
- A sufficient quantity of absorbent material, to soak up any spilt liquids, and suitable containers for the collection and safe disposal of the contaminated absorbent, should be provided at the store.
- Signs should be positioned on or near the store stating, for example, 'Highly flammable' or 'Flashpoint below 32°C'. All signs should conform to the Health and Safety (Safety Signs and Signals) Regulations and *Safety signs* (BS 5499).
- Naked flames, smoking, matches or lighters must not be allowed in the store, and signs stating this must be clearly displayed.

- The use of mobile phones in, or adjacent to, the store should be prohibited. Notices to this effect should be provided and prominently displayed.
- Any lighting or other electrical apparatus must conform to BS EN 60079-14.
- Any shelves or racking in the store should be of a non-ferrous metal or other non-combustible construction.

9.7 Handling and use of dangerous substances

The use of any dangerous substance, including decanting small quantities for daily use from bulk containers, must be the subject of a risk assessment.

The findings of the risk assessment will indicate the maximum quantity of the dangerous substance that can be taken to the place of work and the safe working practices to be observed once it is there and being used.

Generally, only enough of the dangerous substance to enable the work in hand to be carried out should be taken to the place where it is to be used.

The HSE recommends that only the minimum quantity needed for frequently occurring work, or that required for use during a half day or one shift, should be present in the workroom/working area.

Clearly, actual quantities will depend on the work activity and also the organisational arrangements for controlling the fire risks in the workroom/working area.

Decanting, mixing or sampling should not be carried out in a store. It should be done in the open air or in a separate room constructed of fire-resisting materials.

Funnels or pumped nozzles with automatic cut off should be used to prevent spillage whilst decanting is taking place and drip trays should be used to catch any spillage that may inadvertently occur.

Any spillage should be soaked up using proprietary absorbents, dry earth or dry sand.

Metal bins with lids should be provided for any used absorbents to be placed in and these should be clearly identifiable with signs or colour coding and emptied regularly and carefully.

Handle flammable liquids with care

Consideration must be given to the disposal of any waste as it may well be classified as hazardous waste. Such waste must be transported by a registered carrier to a licensed landfill site. Depending on how much waste is produced, the relevant regulator may need to be notified: the Environment Agency (England), Natural Resources Wales, Northern Ireland Environment Agency or the Scottish Environment Protection Agency.

In general, where work involves the use of a dangerous substance that has the potential to create an explosive atmosphere inside a room, all electrical power should be turned off unless all electrical fittings are intrinsically safe by design. If space heating is needed, it should be flameproof and incapable of causing the ignition of any vapours present in the atmosphere.

The build-up of concentrations of vapours must be avoided and dispersed, if necessary, by natural or mechanical ventilation. If mechanical ventilation is necessary, a flameproof motor, not in the ventilation trunking, should be used.

Other points to be considered.

- Naked flames, welding and heating torches, and smoking materials should be prohibited in any area where an explosive atmosphere may be present.
- Metal bins with lids must be provided for offcuts, waste or rags, and should be emptied regularly.
- A suitable container with a lid should be used for any brushes or scrapers that require soaking to remove residues of dangerous substances. This should be placed in a safe area well away from any possible source of ignition.

9.7.1 Spraying of dangerous substances

New covering materials (such as paints, varnishes and lacquers), and the techniques for applying them, have been developed, and extensive use of spray painting equipment can now be seen on construction sites.

Using a spray gun for spraying dangerous substances is most likely to introduce an airborne explosive mist into the workplace, creating a hazard both to the user and to other workers in the area. Such work must only be undertaken by fully trained and competent employees and in a situation where all appropriate precautionary measures are in place.

A risk assessment must be carried out and other controls, including permits to work and/or permits to enter (for those people involved in the job, by implication, excluding all others), implemented as necessary.

Other points to be considered:

- Identify the material carefully and always follow the manufacturer's instructions on preparation, use and application.
- Always use the correct type of spraying equipment. Never make do, just because the proper equipment is not immediately to hand.
- If alternative control measures are not available or adequate, protective clothing and respiratory protective equipment (RPE) must be used.
- Always use the hygiene and washing facilities provided.
- Do not introduce ignition sources into the working area.
- Do not smoke or use naked flames in the working area.
- Always place warning signs in approaches to the area where the work is being carried out, and at entrance points to areas where dangerous substances are being used. Use physical barriers to stop unauthorised persons entering the area.

9.7.2 Empty tanks and containers

Do not cut or heat any empty tanks, containers or drums unless they have been certificated as being free of flammable vapours that could explode. Current opinion recommends reducing the length of time that such certification is valid for. Under most circumstances, the cutting work should be planned to start as soon as the gas-free certificate is issued.

Special care is necessary when demolishing or dismantling disused bulk tanks. By disturbing the tank or heating the residues left inside, you may cause an explosive concentration of vapours. It is important to be aware that even cold-cutting techniques (such as hydraulic shears) may cause sparks that lead to explosion.

Entry into any disused tank or vessel that may be regarded as a confined space should be avoided by doing the work from outside, if possible. Often, there will be a need to clean residues and if entry to a confined space is unavoidable, a safe system of work must be followed and the work carried out under a permit to work system.

Further advice on permit-to-work systems and entry into confined spaces is contained in the Approved Code of Practice (ACoP) and guidance notes that support the Confined Spaces Regulations 1997 (L101) and HSE leaflet *Confined spaces: a brief guide to working safely* (INDG258).

There are a number of factors to consider when the work involves large tanks. The first would be what the contents were. Oil storage tanks may have held so-called heavy fuel oil and these will undoubtedly have been insulated. It is quite likely that the insulation system will have been asbestos.

If the tanks once held petrol, it may have been leaded fuel. This means that exposure to asbestos fibres or lead fumes during cutting should be considered and suitable precautions taken in relation to the presence of asbestos or lead.

It is normal for large tanks, whether above or below ground, to be emptied and cleaned by a specialist contractor before dismantling. The contents are generally removed by a large vacuum tanker and then the inside of the tank is steam cleaned. Most of this work can be carried out from the outside and it is only to carry out the final clean that entry is required.

Operatives carrying out this final clean must be trained in confined space working and be provided with all the normal gas detector, rescue equipment and PPE that would be expected for confined space working.

Having been cleaned, the tank is tested and a gas-free certificate issued. It should then be cut up as soon as possible. The risk of not doing so is that it is practically impossible to completely clean a tank, particularly where its construction incorporates internal ribs, welds and other internal features that could harbour residue of the content.

These residues may become fumes, and if the concentration becomes high enough then the atmosphere inside the tank may become explosive if ignited.

9.7.3 Industrial gas cylinder colours

While the cylinder label is the primary means of identifying the properties of the gas in a cylinder, the colour coding of the cylinder body provides a further guide.

The colour applied to the shoulder, or curved part at the top of the cylinder, signifies the European Standard colour coding.

The aim of the new standard (EN 1089-3) was to replace the old cylinder colour scheme (BS 349), to help improve safety standards within the gases industry.

www **_Gas cylinder identification. Label and colour coding requirements_ (TIS6) can be downloaded for free from the British Compressed Gases Association website.**

Operators should obtain a cylinder identification chart from their supplier, as different gas suppliers may paint the body of a cylinder different colours. For example, a BOC oxygen gas cylinder has a black body with a white shoulder whilst an air products cylinder has a grey body with a white shoulder.

9.8 Liquefied petroleum gas

LPG is a mixture of hydrocarbons that are a gas or vapour under normal conditions of temperature and pressure, but can be turned into a liquid by either the application of pressure or the reduction of temperature.

LPG can be found in numerous locations, in various sizes of cylinder, and can be put to a variety of uses on building and construction sites. Uses range from the heating of bitumen boilers, site huts and offices to providing a fuel for hand tools and cutting equipment.

If used properly and safely, LPG is a convenient and valuable source of energy. Misuse or carelessness can cause serious accidents.

LPG and commercial butane and propane

Liquefied petroleum gas. Means any commercial butane, commercial propane or a mixture of the two.

Commercial butane. This is usually stored in blue cylinders. It consists mainly of butane and butane isomers. The remaining components are predominantly propane and propane isomers, pentane and pentane isomers. Because of the low vapour pressure, butane cylinders are not generally used outside.

Commercial propane. This is usually stored in vessels or in red cylinders. It consists mainly of propane and propane isomers. The remaining components are predominantly butane and butane isomers, ethane and ethane isomers.

Important points about LPG

- In some cases, the Regulatory Reform (Fire Safety) Order will only be relevant where LPG is stored off site (for example, at company premises and builders' yards). However, it will also apply to some construction sites where LPG is used in conjunction with a work process.
- LPG is normally found as a compressed liquid, usually of commercial butane or propane.
- LPG is a colourless, odourless liquid that floats on water but vaporises to form a gas that is heavier than air. A stenching agent is normally added.
- A release or spillage of LPG can form a large vapour cloud of flammable gas capable of ignition from some distance.
- LPG is stored on site in fixed tanks, refillable cylinders or non-refillable disposable cylinders (cartridges).
- Storage should be in secure, non-combustible, well ventilated areas away from other risks and sources of ignition.
- There are special requirements for vehicles and drivers covering the transportation of LPG.
- In site huts all LPG cylinders and regulators for use with fixed heaters, cookers and lighting must be kept outside and piped in using rigid copper piping.
- Staff who work with LPG must be suitably trained in the hazards and use of LPG (such as not rolling cylinders).
- In the event of a leak, do not attempt to operate electrical apparatus or switches.
- If a fire breaks out that involves LPG cylinders:
  - immediately inform the fire and rescue service of the whereabouts of all cylinders on site, including details of whether they are full or empty
  - if in any doubt as to the safety of the overall situation, evacuate the site and put a security cordon in place.

9.8.1 Properties

As a liquid, LPG is lighter than water and will float before evaporating.

As a gas, it is approximately twice as heavy as air and will sink and flow into sumps and underground excavations or workings. It will also sink into drains but, because its density is approximately half that of water, it will not flow through drains that are water-trapped.

As a gas, it is capable of ignition at some distance from the original leak. The resulting flame can travel back to the source of the leak.

Any release of liquid under pressure into the atmosphere results in its rapid conversion to gas. This gas has a volume of about 230 (butane) and 270 (propane) times that of the liquid.

The expansion, during a rapid release of pressure, results in a rapid drop of temperature, which for propane can approach its boiling point of -45°C. Leakage of liquid LPG will result in the release of large volumes of highly flammable gases.

Property	Commercial butane	Commercial propane
Density in comparison to water.	0.57	0.5
Density in comparison to air.	2	1.5
Litres per tonne.	1,745	1,995
Boiling point.	-2°C	-45°C
Pressure at 15°C.	1.5 bar	7.0 bar
Expansion ratio.	1:230	1:270
Levels of flammability.	1.9%–8.5%	2.0%–10.9%

For example, one litre of liquid propane spilt in a workplace will evaporate to make approximately 270 litres of gas. If it is diluted with air to 2%, this will give 13,750 cubic litres of an explosive gas/air mixture – enough to fill a room 3 m x 2.3 m x 2 m.

The use of LPG equipment in confined spaces, and small, poorly ventilated spaces (such as basement and sub-basement boiler houses, toilets, site cabins and kitchens) can give rise to a highly flammable or explosive atmosphere, if the equipment should leak.

9.8.2 Flammability

Following mechanical failure of LPG equipment, or any other event that causes the release of LPG, the resulting gas will form a flammable mixture with air at gas concentrations between approximately 2% and 10%.

Ignition of released LPG, where the concentration exceeds 2%, can result in fire or, if confined, an explosion. If a leak does not ignite immediately, and the LPG and air mixture drifts from the point of release, it will gradually become more diluted.

However, should the concentration still exceed 2% and ignition occur, this could cause a flash or cloud-fire back to the point of release.

A leak of LPG may sometimes be noticed either by the smell or the noise of the gas escaping. There may also be condensation or frosting on the outside of the cylinder.

All LPG systems must be leak tested each time prior to use. An approved leak detection fluid (not soapy water) should be applied to all joints and connections. The presence of bubbles indicates that the system is not gas tight and should not be used. If a leak is discovered the system should be shut down, safely de-pressured and the leak rectified. LPG leaks must never be traced with a lighted match or flame.

If it is suspected that LPG has leaked inside a building, no attempt should be made to touch any electrical apparatus. Do not turn light switches, sockets or any other electrical appliance either on or off. Open all doors and windows, if it is safe to do so, and leave immediately. Do not re-enter the building until advice has been sought and you are told that it is safe to do so.

9.8.3 Health issues

9.8.3.1 Workplace exposure limits

The maximum levels of exposure, as stated in EH40/2005 workplace exposure limits under the heading LPG, are:

- 1,000 ppm (0.1%) for long-term exposure (reference period eight hours)
- 1,250 ppm (0.125%) for short-term exposure (reference period 15 minutes).

During any maintenance work involving release of pressure, especially in confined spaces, care must be taken that these exposure limits are not exceeded.

9.8.3.2 Inhalation

LPG gas is not toxic, but at concentration levels above about 10,000 ppm (1%) in air, propane becomes a slight narcotic. At higher levels, it becomes an asphyxiant by displacing oxygen.

In a sufficiently high concentration, a person will suffocate and die.

9.8.3.3 Cold burns

The release of liquid propane onto unprotected skin will cause cold burns. This is due to the rapid vaporisation of the liquid, withdrawing heat from the affected area of the body.

The release of liquid, or significant amounts of gas at vessel pressure, can also cause the adjacent fittings to cool. This may be sufficient to cause cold burns if the fittings are touched by unprotected hands.

Suitable skin and eye protection must be worn whenever there is the possibility of a release of liquid LPG.

In the event of a cold burn, treat as for a burn from a hot object. The burn area should be placed under cool running water for 20 minutes and you should seek first aid or professional medical help (call 999). Be aware, shock may occur due to intense pain.

9.8.4 Environmental hazards

A small, unignited release of LPG would not pose a serious danger to the environment.

The gas, being heavier than air, will roll and sink to the lowest point (such as a basement or excavation). LPG vapours can travel large distances and this may result in a fire or explosion, even if the source of ignition is a considerable distance from the original leak.

A fire and explosion would be instantaneous on ignition and would be limited to immediate damage. The fire might devour only escaping LPG and then the danger will have passed with no lasting environmental damage.

The fire will burn fast and the explosion will be intense, but both may be over very quickly.

9.8.5 Storage

LPG can be stored on construction sites in one of three ways.

- Fixed storage tanks.
- Refillable cylinders.
- Non-refillable cylinders (that is, disposable cylinders).

9.8.5.1 Fixed storage tanks

Whilst most LPG used on construction sites can be found in cylinders, on some larger sites there may be a need for bulk storage. In view of the large capacity, it is essential that the positioning of any storage tank is carefully planned and discussed with the local fire prevention officer and the HSE.

LPG tanks should be positioned on a level concrete base to provide a stable foundation. For short-term installations it may be satisfactory to stand the tank on concrete slabs, but advice must be sought from the tank or gas suppliers. Consideration also needs to be given to any pipework that may run from the tank to the application.

Tanks should not be sited close to any ditches, cellars or drains, and delivery and emergency vehicles must easily reach them.

If the bulk tank is used as a supply vessel for filling LPG cylinders on site then reference should be made as to how this procedure can be undertaken safely.

All access roads must be clear of obstruction and the entire area kept free from weeds and other vegetation.

Tanks over 2,250 litres liquid capacity should be electrically bonded and earthed.

All bulk storage tanks must have good, all-round ventilation. On non-secure sites, tanks should be protected against vandalism by a chainlink fence at least 2 m high.

Motorway-type crash barriers should surround the installation to minimise damage by motor vehicles.

Installations must be clearly labelled 'Highly flammable LPG. No smoking or naked lights'.

Signs must conform to the Health and Safety (Safety Signs and Signals) Regulations and *Safety signs* (BS 5499). Signs purchased from reputable suppliers will be to these specifications.

Adequate separation must be maintained between bulk storage tanks and adjacent buildings or boundaries. As a guide, the separation distances for bulk LPG tanks, detailed in the table on the right, should be followed.

Where possible, LPG storage areas should not be positioned under power cables. Where this is unavoidable, the **minimum** distances between the extremities of the vessel or cylinders to the nearest cable should be:

- up to 1 kV – 1.5 m
- 1 kV or above – 10 m.

Gas capacity	Water capacity		Minimum distance*
Tonnes	Litres	Gallons	Metres
Under 0.2	450	99	2.5
0.2-1.0	451-2,250	100-495	3
1.1-4.0	2,251-9,000	496-1,980	7.5

** Minimum distance from boundaries, buildings or sources of ignition.*

The above figures are for guidance only. Advice should always be sought from the power supply company before any work takes place in the vicinity of overhead power cables.

9.8.5.2 Refillable cylinders

A level base of compacted earth, concrete or paving slabs should be provided and surrounded by a secure chainlink fence at least 2 m high. A hard standing should be provided for the delivery and dispatch of cylinders. The area should be kept weed and vegetation free.

If the compound is more than 12 m square, two exits should be provided in opposite corners of the compound. The emergency exits should be fitted with a panic bar so that they can be easily opened in an emergency. If it is less than 12 m square, one gate will suffice.

Gates should open outwards and always be left unlocked when someone is in the compound. There should be sufficient shelter to prevent cylinders from being exposed to extremes of weather.

Signs must be clearly displayed indicating the presence of LPG, and prohibiting smoking and the use of any naked flame in the area of the store.

LPG cylinders must be stored with their valves uppermost. They must be stored away from oxygen, highly flammable liquids, oxidisers, toxic or corrosive gases or substances. A distance of at least 3 m must be kept between LPG cylinders and other such substances, although they may be kept in the same compound.

Any store for refillable LPG cylinders must be located away from boundaries, buildings, fixed sources of ignition or electrical equipment by at least the distances detailed in the table below.

LPG storage (including empties)	Separation from building/boundary
Below 50 kg	1 m
50–1,000 kg	3 m
1,001–4,000 kg	4 m

The store must be sited at least 3 m away from any cellars, drains or other excavations into which a leak of gas would collect. If only a small compound is used (3 m² for example) cylinders may be stored against the inside of the compound fencing, providing this fence is not within 3 m of any boundary.

Empty cylinders must be stored with their valves securely closed to prevent any residue of gas escaping, or air being drawn into the cylinder. Full and part-used cylinders must be stored separately from empty cylinders in properly identified areas.

Stocks should be grouped in batches of not more than 1,000 kg and batches separated by a minimum 1.5 m gangway. However, if there is less than 50 kg of LPG these may be stored with other gases. Where lighting is necessary, it should be mounted well above ground level and not less than 2 m above the cylinders.

Any equipment not in use (such as portable hand-held equipment) should be isolated so as not to be easy to get to for trespassers. Any cylinders not required should be returned to the storage compound or other secure position.

9.8.5.3 Non-refillable cylinders

Non-refillable LPG cylinders for use with small portable equipment (such as blowlamps) may be stored in a clearly identified, lockable, metal container, adequately ventilated, to prevent a build up of LPG gas. Care should be taken when changing cylinders to ensure that connections are correctly made and that there are no leaks. Always dispose of empty containers safely and in accordance with the manufacturer's recommendations. Do not, under any circumstances, puncture or throw empty cylinders onto a fire.

Small refillable cylinders (for example, primus) should also be stored in the same way as non-refillable cylinders. Although only containing small quantities of gas, they must not be stored in occupied site huts. They should be kept in a secure, non-combustible, well-ventilated external enclosure. The store should have warning signs 'Highly flammable – LPG', and prohibition signs 'No smoking/naked lights'.

The disposal of cartridges after use requires care as they still contain gas. Under no circumstances should cartridges be thrown in general waste skips or on fires. All empty cartridges and non-refillable cylinders should be disposed of by a specialist waste contractor. Small refillable cylinders should be kept and returned to the supplier when empty.

9.8.6 Cylinders

9.8.6.1 Handling

Care must be taken when moving cylinders around the site, especially by hand or on rough ground. A full 47 kg cylinder has a total mass of about 90 kg and, before moving by hand, requires an assessment under the Manual Handling Operations Regulations 1992 of the method to be used. Cylinders must not be rolled, even when empty.

Cylinders should be handled with care and, wherever reasonably practicable, moved using suitable equipment. They should not be moved unprotected in dumper trucks or on forklift trucks. The valve on a cylinder should not be used for lifting or to lever the cylinder into position. Damage to the valve can result in a non-controllable release of LPG under high pressure. Throwing cylinders from any height or dropping them is prohibited, as in such circumstances damage to the valve, shroud and cylinders is even more likely.

9.8.6.2 Damaged cylinders

Before use, cylinders should be examined. Any damaged or faulty cylinder should **not** be used. The cylinder should be labelled and the supplier immediately requested to collect the cylinder. If a cylinder is found to be leaking (usually from the valve) and the leak cannot be stopped, the cylinder should be carefully removed to a well-ventilated, open space, free from sources of ignition. It should be left with the leak uppermost, marked faulty, and notices displayed prohibiting smoking or other naked lights.

The area where the leaking cylinder is situated should be cordoned off with barriers to prohibit access. The supplier of the cylinder and, if necessary, the FRS, should be informed immediately. If it is necessary to inform the FRS, it may be good practice to evacuate any personnel to a place of safety, away from the leaking cylinder.

Under no circumstances should attempts be made to dismantle or repair defective cylinders.

9.8.6.3 Orphaned compressed gas cylinders

Whilst this advice applies to abandoned LPG cylinders, it is also valid for compressed gas cylinders that have held other gases.

Empty compressed gas cylinders may still contain some of their original content. They are abandoned for a variety of reasons and then appear at metal recyclers or council waste disposal sites, neither of which welcomes their presence.

It is thought that cylinders are often abandoned for the following reasons.

- They are empty and too troublesome to return to the supplier.
- They are damaged and therefore not returnable.
- The owner or supplier cannot be identified.
- They are at the end of their useful life.

Within the UK there is a cylinder retrieval system co-ordinated by the Liquid Petroleum Gas Association. Four major companies own around 90% of all UK compressed gas cylinders. Each company has arrangements in place to collect their own cylinders.

For details of collection companies refer to Appendix A.

9.8.7 Carriage of dangerous goods

This section details the requirements of legislation for people who occasionally carry LPG cylinders in a van, truck or similar vehicle. This is covered by the CDG, which implements ADR.

9.8.7.1 Transportation

Acetylene, LPG and other gases are commonly carried by tradespersons (such as welders, plumbers and motor vehicle repair technicians). Flammable gases are in transport Category 2. Oxygen and gases (such as carbon dioxide and argon) are in transport Category 3. Usually small load threshold exemptions will apply. The parts of ADR which apply are shown below.

- The driver will have received general awareness, function-specific and safety training (ADR 1.3). A record of such training must be kept.
- The vehicle must be equipped with at least one 2 kg dry powder fire extinguisher, which is kept in good working order.
- The load must be properly stowed.

Note that special provisions CV 9, 10 and 36 all apply in the case of these gases. In particular, CV 36 specifies:

… packages shall preferably be loaded in open or ventilated vehicles or open or ventilated containers. If this is not feasible and packages are carried in other closed vehicles or containers, the cargo doors of the vehicles or containers shall be marked with the following, in letters not less than 25 mm high: 'WARNING: NO VENTILATION, OPEN WITH CAUTION'.

The HSE has published a leaflet *Working safely with acetylene* (INDG327) and the British Compressed Gases Association (BCGA) has a useful leaflet *Carriage of small quantities of gas cylinders on vehicles* as well as a wider range of publications that may be helpful.

9.8.7.2 Small load exemptions (ADR 1.1.3.6)

Small load exemptions relate to the total quantity of dangerous goods carried in packages by the transport unit (usually the van or lorry, but also any trailer). It is the transport category (TC) that determines the load limits (thresholds). Many substances are assigned a packing group, but these are not synonymous in all cases with the TC. The TC is given in Column 15 of Table A in ADR (Chapter 3.2). If this is not available, the table at ADR Part 1.1.3.6.3 needs to be consulted.

References to load limits for the different transport categories are given in the table below. For convenience this has been amended in accordance with Regulation 19 but it needs to be used with care.

Vehicle/load	Driver training	ADR reference
All vehicles except those carrying packages under the load limit.	General training plus ADR training certificate. The certificate may be endorsed for different classes of dangerous goods or different modes (in tanks or other than tanks).	8.2.1
Any vehicle carrying packaged dangerous goods under the small load limit.	General training.	8.2.3 *(refers to Chapter 1.3).* ADR 1.1.3.6
Vehicle with small tank (up to 1 m³).	General training.	8.2.1.3 8.2.3

If a vehicle is carrying dangerous goods under the small load threshold, many of the requirements of ADR are not applicable. Some care needs to be taken, as 'what is not exempted is still required'. In most cases the remaining obligations are as follows.

- General training for driver (ADR 1.3.2). A record of the training should be kept (ADR 1.3.3).
- Carry at least one 2 kg dry powder fire extinguisher or equivalent (ADR 8.1.4.2).
- Stow the dangerous goods properly (ADR 7.5.7).

Note that use of these exemptions is optional. For example, a carrier may choose to display the orange plates as long as the vehicle is carrying dangerous goods. All vehicle marks (orange plates) must be removed when dangerous goods are not being carried. An important aspect is that packaging has to comply with the relevant standards.

e.g. Small load application

- LPG. This is in transport Category 2. The small load threshold is 333 kg and LPG is LQ0. (Limited quantity (LQ) exemptions mainly apply to small receptacles, typically of the sort that go into the retail distribution chain, which are packed in boxes or on shrink-wrapped trays.) The result is that all cylinders count towards the load limit, but if that is less than 333 kg, the minimum ADR requirements apply.
- Section 8.2 of the ADR details mandatory training for drivers. There should be a record of their training, which must be carried at all times. The level of training required depends upon the load.

9.8.7.3 General transportation of LPG cylinders

When loaded onto vehicles, cylinders must be kept upright and secured. Vehicles must be equipped with two dry powder extinguishers (nominally a 6 kg and a 2 kg) and a first-aid kit, and must also display warning notices. LPG should not be carried with other flammable substances (such as paints and solvents). If this is unavoidable, such materials should be kept in a steel chest away from the cylinders.

Drivers carrying more than two LPG cylinders must have received training in what to do in an emergency and carry a transport emergency card containing details of the load carried. If a cylinder leaks during a journey, close the valve immediately. If this is not possible, move the vehicle to open ground, away from buildings and people, and inform the emergency services.

www **For information on fire extinguisher requirements when carrying dangerous goods refer to the HSE guidance.**

9.8.8 Safe use and handling of LPG

Everyone with any responsibility for the storage and transportation of LPG must understand the characteristics and hazards of the LPG product they are using. They should be suitably trained and understand the fundamentals of fire-fighting and control of leakages. They should also have knowledge of the procedures for dealing with emergencies. It is not possible to cover all aspects of the use and application of LPG, but the notes below (which should not be regarded as exhaustive) give the main points for its safe use and handling.

Safe use and handling of LPG

- Never use or store a gas cylinder on its side, unless it is a special cylinder for use on LPG-fuelled plant and vehicles. Liquefied gas may escape, causing concentrations of gas, and operatives may suffer frostbite because of the low temperature of escaping liquid.
- Propane cylinders must never be stored indoors because any leakage will lead to large concentrations of explosive mixtures.
- Only hoses suitable for use with LPG installations or appliances should be used and these should be inspected frequently for wear. Hoses fitted with worm drive screws should not be used.
- Cylinders must not be dropped during handling, nor brought into violent contact with other cylinders or adjacent objects.
- After use, valve protection caps and plastic thread caps or plugs should be fitted to prevent accidental leakage.
- LPG cylinders should not be used below ground level as any leakage of gas will collect at the lowest point and will not disperse.
- Regulators must be handled with care, and should not be used if damaged, but replaced or sent for specialist repair. The regulator should be labelled for the type of gas it is designed for. It should be CE marked, comply with European Standards and be in date.
- Hoses and fittings should be examined before use. Damaged items must be replaced.
- LPG cylinders are fitted with a left-hand thread or push-on connection. Union nuts and couplers have grooves on the outside corners of the nuts confirming this. Always use the correct size spanner to tighten or loosen connections. Hand-tight connections will permit leaks. Over-tightening will damage threads and cause leaks.
- Checks for leaks should be carried out using a proprietary leakage detector. **Never use a match or other naked flame.**
- Before connecting any LPG cylinder to equipment, it is essential that all fires, flames or other potential ignition sources, including smoking materials, are extinguished. Where reasonably practicable to do so, cylinders should be changed in the open air.
- If a leak is found, the gas supply must be turned off at the cylinder immediately.
- Flexible hoses should be in good condition and protected or steel braided if they are likely to be subjected to damage by abrasion. Hoses must conform to BS EN 16436, *Rubber and plastics hoses, tubing and assemblies for use with propane and butane.*
- Before use, inspections should be carried out on all LPG appliances and equipment. The inspection should cover testing for leaks and the cleaning, adjusting and checking of hoses, hose clips and ferrules.
- Empty cylinders should always be treated with the same caution as new ones and be returned to a properly designated central storage area for collection. Under no circumstances should an LPG cylinder, either full or empty, be left around the site or buried during site operations.

9.8.9 Regulators

LPG regulators should be suitable for the equipment with which they are to be used. They should be suitable for either propane or butane and be set to the correct pressure. They should be capable of passing the correct flow capacity. It is dangerous to use regulators set at the incorrect pressure. Cylinders should not be transported with regulators and hoses attached, unless on a purpose designed trolley or carrier.

Before assembling regulators and fittings ensure that the cylinder valve outlet is clean, dry and free from damage and dirt. This will help to prevent the ignition of components. Oxygen regulators should be kept free from oil or grease and be suitable for the maximum cylinder pressure being used. Oxygen cylinder valves should be opened slowly.

Before attaching a regulator to a cylinder the following checks should be made.

- The regulator is within its expiry date (all regulators have a lifespan, on expiry of which they require either replacement or refurbishment).
- The gas inside the cylinder is correctly identified and the regulator is suitable for that specific gas.
- The maximum cylinder pressure is identified.
- The regulator is suitable for the maximum cylinder pressure (regulator inlet pressure).
- The regulator has a suitable outlet pressure for the application.
- The regulator is in a serviceable condition.
- The gauges are not damaged and do not show signs of over pressurisation.
- The cylinder valve outlet thread is mechanically compatible with the regulator inlet connection and is clean and free of dirt. A lint free cloth can be used to clean the outlet.
- The regulator outlet thread is in good condition.
- The regulator has the manufacturer's or supplier's name clearly visible.
- The regulator can be fitted at a suitable orientation.
- The regulator pressure adjusting screw is set to zero pressure position by turning the control knob fully anti-clockwise.
- The correct sized spanner is available for securing the regulator.
- The inlet connection should be inspected for damage. If an 'O' ring is fitted to the inlet, check for damage and replace, if necessary, with an 'O' ring recommended by the regulator manufacturer.

Jointing paste or tape should never be used between the regulator and cylinder valve.

9.8.10 Bitumen boilers and cauldrons

Work with a bitumen boiler should always be supervised by a competent person.

The majority of boilers or cauldrons are fuelled by LPG to melt the block bitumen. Such a boiler or cauldron must be sited on a level, non-flammable base, away from pedestrian routes and areas where site traffic may damage hoses or gas cylinders. If avoidable, bitumen boilers should not be taken onto roofs.

Sequence for lighting	
1.	Remove the burner from the boiler or cauldron.
2.	Have the source of ignition ready before turning on the gas.
3.	Light the burner, ensuring that the gas is on slowly.
4.	Replace the burner beneath the boiler or cauldron.

Ensure that any LPG cylinder is at least 3 m away from the boiler or cauldron to which it is attached. Full cylinders, not attached, should be kept at least 6 m away from the boiler or cauldron and protected from heat. Supply hoses should be checked for crushing, damage to the metal braiding or impregnation with bitumen. Any unserviceable hose must be replaced.

In some circumstances (for example, due to levels of atmospheric humidity (climate conditions) and when using gas rapidly) frost, condensation or ice can form on gas bottles and/or gas regulators. The propane itself would not freeze, as that would require temperatures below the freezing point of -188°C.

If you are running a boiler or cauldron and frost forms on the outside of the cylinder, it may indicate that the gas flow rate is too high. Either use a smaller burner, to regulate the flow, or couple two or more cylinders together, by means of a manifold, to supply the larger burner at a suitable flow rate. Never leave a bitumen boiler or cauldron unattended when the burner is alight, and never move a bitumen boiler or cauldron with the burner alight. Always ensure the boiler or cauldron and bitumen has fully cooled before attempting to move it.

If a bitumen boiler or cauldron is overfilled, overflows or boils over, the LPG cylinder valves must be turned off immediately. Any spillage should be contained using dry sand or earth and then left until cool. No attempt should be made to remove or recover any spillage of hot bitumen. Where bitumen boilers are located at ground level, additional protection measures must be in place to prevent the public from gaining access to the boiler. If boilers are located on scaffolds they must offer protection to the public from spilt bitumen. This can be achieved with additional boards and edge protection. A dry powder extinguisher, of a minimum 4.5 kg in size, should be provided whenever a bitumen boiler or cauldron is used.

9.8.11 Gas-operated hand tools and equipment

There are two types of LPG cylinder available for use with portable tools: disposable and refillable.

These cylinders come in various shapes, sizes and colours, depending on the manufacturer. They range in size from small (0.5 kg) to large (47 kg).

All LPG cylinders used with portable equipment should be positioned upright and secured (if possible). Cylinders used with cutting equipment should always be placed on purpose-made trolleys.

- Before changing a cylinder, always make sure that all valves are closed.
- Hoses must never be kinked to try to shut off gas when changing torches. It does not work and can lead to a gas escape.
- Always replace valve protection caps and plastic thread caps.
- Flames from portable tools must not be allowed to play on LPG cylinders.
- When work has been completed, turn off the cylinder valves and allow the flame from the portable torch to burn out.
- Closure of torch valves rather than cylinder valves will retain gas in hoses which, if damaged, will allow gas to escape.
- Hoses and torches must never be put into site toolboxes while still attached to the cylinder.
- Manufacturers' operating pressures must be strictly observed and must never be exceeded.
- Do not interfere with preset pressure regulators.

9.8.12 Gas-powered fixing tools

Many of the principles for the safe use of cartridge-operated tools also apply to gas-powered fixing tools, which use a canister of pressurised gas (a fuel cell) as a propellant.

Generally, gas-powered fixing tools are used for firing fixings into softer materials (such as timber). However, in untrained hands they can be as dangerous as cartridge-operated tools.

The implications of a misfire when using a gas-powered fixing tool are not as serious as when using a cartridge-operated tool and it is usually safe after a misfire to attempt to make the next fixing immediately. The battery and fuel cell must be removed prior to attempting to remove a blockage.

However, it is important to store and dispose of canisters safely *(refer to 9.6 Storage of dangerous substances).*

9.8.13 LPG for use in site huts and other small buildings

All LPG cylinders and regulators for use with fixed heaters, cookers and lighting within site huts must be kept outside and the gas supply piped in using rigid copper piping.

The use of flexible hosing is permitted only between the cylinders and changeover valves or manifolds, and for the final connection to appliances, but this must be kept as short as possible.

All pipework should be exposed and easy to get to for inspection, but located to prevent accidental damage. Any work on LPG pipework or other parts of a fixed installation, including testing, must only be carried out by appropriately trained (Gas Safe registered) persons.

Ventilation for heaters and cookers must be permanent and adequate. It should be divided equally between vents at high and low level.

A two-burner cooker in a site hut needs approximately 150 mm x 150 mm ventilation. A 3 kW convector heater needs approximately 225 mm x 225 mm ventilation.

Inspections of all appliances must be carried out before use. If soot forms or smells occur do not use or allow the appliance to be used. Find out the reasons for the problem and ensure that it is corrected.

9.8.14 Enclosed spaces

Before using LPG equipment in an enclosed space, it is essential to carry out a risk assessment under the Management of Health and Safety at Work Regulations.

It is essential to ensure that there is adequate ventilation, which may have to be forced. This is necessary to ensure full combustion and also to make certain that the products of combustion, other fumes and excess oxygen from any cutting apparatus are removed. Proper safety precautions and atmospheric monitoring should be considered.

Wherever practicable, cylinders used with operations in confined spaces should be located in a safe area, preferably in the open air. The supply pressure should be reduced to the lowest practicable level on leaving the source of supply.

Where cylinders are used below ground level, the number must be kept as small as possible.

All cylinders and hoses should be removed as soon as work has finished or if it is interrupted for a substantial period (for example, overnight).

LPG cylinders must not be taken into confined spaces, as defined in the Confined Spaces Regulations, unless exceptional safety precautions are taken.

9.9 Fire

Fires involving flammable liquids usually fall into one of two categories.

- Flowing liquid fires.
- Contained liquid fires.

Powder extinguishers are the most suitable type for tackling a flowing liquid fire. The use of foam or carbon dioxide extinguishers may be effective on a small, flowing liquid fire.

Foam extinguishers are the most suitable type for use on contained liquid fires. Powder or carbon dioxide extinguishers may also be used, but operators should be aware of the short duration of small carbon dioxide extinguishers and the possibility of re-ignition of any residual vapours being given off when an ignition source is still present.

Do not use a water extinguisher for any fire involving flammable liquids. The water will not extinguish the fire, but will cause a violent reaction resulting in the burning liquid spreading around much faster. This could spread and intensify the fire.

Suitable portable fire extinguishers should, wherever possible, be sited in pairs (so as to minimise the risk of failure) and in strategic positions adjacent to the store.

9.9.1 Colour of fire extinguishers

With the introduction of the British Standard *Portable fire extinguishers* (BS EN 3) there are now two colour coding systems for portable fire extinguishers.

It is important that staff who may be called upon to use a fire extinguisher have a clear understanding of the colour coding.

This standard states that the body of all fire extinguishers should be red with zones of colour covering not more than 5% of the surface area of the extinguisher, fixed to the extinguisher body and denoting the extinguishing agent or medium it contains.

Colour coding by agent or medium (see right) enables a trained person to rapidly identify the type of extinguisher needed in an emergency.

Other information concerning its use may also be displayed on the body of the extinguisher.

Extinguishing medium	Colour of panel
Water	Red
Foam	Cream
Powder (all types)	Blue
Carbon dioxide	Black
Wet chemical	Yellow

The requirements of the standard are not retrospective. Fire extinguishers conforming to the older standard, BS 5423, which has been withdrawn, will continue to be found on premises and they can remain in use and be refilled until they need to be replaced. When replaced, the new extinguishers must conform to BS EN 3.

The full body colour coding of the older extinguishers is the same as is listed in the table above. In addition, a small number of extinguishers are bright silver or self-coloured metal with a designated panel stating the medium they contain. The colour coding of the panel is the same as the above.

Training in selecting the correct type of extinguisher to use and the safe way to operate firefighting equipment is essential and should be undertaken by all staff who work with dangerous substances. The use of the wrong extinguisher in the wrong way would have serious consequences.

Advice on training can be obtained from extinguisher manufacturers or the local fire station.

9.9.2 Enforcement of fire safety legislation

The responsibility for the enforcement of fire safety legislation is split between the following organisations.

- HSE, or possibly the Local Authority on small sites, either of which would usually take enforcement action under CDM or the Management of Health and Safety at Work Regulations (MHSWR). However, the HSE also has enforcement powers under the Regulatory Reform (Fire Safety) Order.
- FRSs, which would enforce their powers under the Regulatory Reform (Fire Safety) Order (England and Wales), the Fire (Scotland) Act or the Fire Safety Regulations (Northern Ireland).

In most circumstances, fire safety legislation will be enforced by the following organisations.

- HSE on construction sites, including on-site offices and other on-site accommodation.
- FRSs where work other than construction is taking place, for example where:
  - one floor of an office block is being refurbished but the other floors remain occupied
  - part of a department store sales floor is a cordoned-off construction site but the public have access to the sales floor outside of the hoarding.

In the two circumstances described above it is usual for the HSE to enforce fire safety legislation within the confines of the site and the FRS to enforce it outside the hoardings. However, the responsibility for the enforcement of fire safety legislation may not always be as clear cut as this. For example, the FRS is likely to take an interest in any construction site where there is a significant fire loading or there are other factors that significantly increase personal risk. Some examples are shown below.

- Where the flammable nature of the partly finished structure is considered to pose a significant risk should it catch fire.
- The flammable (or highly flammable) nature and quantities of materials and substances stored and used on the site (for example, LPG).
- The vulnerability of adjacent premises should a fire start on site.
- If there is a large on-site multi-floor administrative complex housing technical staff and welfare facilities.

This aspect of fire safety is considered to be particularly relevant to this chapter if significant quantities of flammable or highly flammable substances are stored on site.

www **For further information on legislation and other resources, including guidance and incident reports, visit the HSE website.**

9.9.3 Liquefied petroleum gas

9.9.3.1 Action in an emergency

Instructions for dealing with incidents involving LPG will vary for each situation. The most important thing is to avoid endangering anyone's life. The following actions should be taken by anyone discovering a fire in the vicinity of LPG cylinders.

- In case of fire, **no matter how small**, call the FRS.
- Whilst waiting for the FRS to arrive, (if it can be done safely) turn off all cylinder valves to cut off the fuel supply and remove the cylinders from the area.
- If this action cannot be completed safely evacuate the site and impose a cordon to stop anyone inadvertently entering the area.
- General advice is to leave tackling any fire involving cylinders or tanks to the FRS unless it is judged that the fire can be put out without endangering anyone.
- Training in the correct type of fire-fighting equipment to use, and the safe way to operate it, should be undertaken by all staff who work with LPG. These staff must be trained to recognise when the situation is getting out of control and know when to evacuate the area.
- When the FRS arrives, inform the fire officer of the situation including:
  - the location and contents of all cylinders
  - details of any security cordon that you have implemented
  - confirmation that all people who were known to be in the area have been accounted for or details of anyone that is unaccounted for
  - if possible and required, offer them the data information sheet relating to the cylinder(s) involved.

9.9.3.2 Remember

- Cylinders fitted with pressure relief valves can produce gas jets that will extend a considerable distance.
- If cylinders are exposed to a severe fire or are engulfed in flames, no attempt should be made to fight the fire. **Evacuate everyone from the area.**
- Where a flame from a leaking gas cylinder is extinguished but the valve is still open, gas will continue to escape and there will be a danger of a gas cloud forming and the risk of an explosion.
- Any cylinder involved in a fire should be clearly labelled that it has been involved in a fire and removed from the area to a safe place. Telephone the suppliers – they will give advice and arrange for the cylinder(s) to be collected.

Instructions concerning emergency procedures should be clearly displayed and employers should ensure that all employees fully understand them. Emergency procedure drills should be practised regularly. Data information sheets are available from product manufacturers giving advice in case of an accident involving LPG cylinders. A copy of each sheet should be available for inspection and those sheets relating to the cylinders involved should be given to the fire officer.

9.9.4 Firefighting equipment

Employees who work with LPG should be identified and trained in the selection and use of firefighting equipment. Portable extinguishers should be used as a tool to preserve life where it is at risk, or a means to help you safely escape a dangerous situation. They should not be used as a tool to actively approach and fight an LPG fire. Advice on the training of staff can be obtained from the LPG supplier or the FRS.

Portable extinguishers, sited in pairs to minimise the risk of failure, should be positioned at strategic points wherever LPG is stored or used. As a guide, no fewer than two 4.5 kg dry powder extinguishers or equivalent should be provided for every 20 large (47 kg) cylinders stored.

9.9.5 Training

Most accidents involving LPG are due to ignorance of basic safety precautions. All persons using LPG cylinders, tools or equipment should be suitably instructed in the hazards associated with LPG, and the precautions to be taken in its use.

9.10 Legislative requirements

9.10.1 Construction (Design and Management) Regulations 2015

These regulations specify the measures to be taken to prevent the risk from fire, explosion or any substance likely to cause asphyxiation and include the measures that should be taken to detect and fight fires, which could occur on a construction site.

In practice, the HSE is likely to use this legislation for any enforcement action relating to the risks mentioned above.

For further information refer to Chapter A03 Construction (Design and Management) Regulations 2015.

9.10.2 ATEX and explosive atmospheres

ATEX is the name commonly given to the two European Directives for controlling explosive atmospheres.

1. Directive 99/92/EC (also known as ATEX 137 or the ATEX Workplace Directive) on minimum requirements for improving the health and safety protection of workers potentially at risk from explosive atmospheres.
2. Directive 2014/34/EU reflects the laws relating to equipment and protective systems intended for use in potentially explosive atmospheres.

In Great Britain the requirements of Directive 99/92/EC were put into effect through Regulations 7 and 11 of the Dangerous Substances and Explosive Atmospheres Regulations (DSEAR).

The aim of Directive 2014/34/EU is to allow the free trade of ATEX equipment and protective systems within the EU by removing the need for separate testing and documentation for each member state.

In Great Britain, the requirements of the directive were put into effect through the Equipment and Protective Systems Intended for Use in Potentially Explosive Atmospheres Regulations (SI 1996/192).

The regulations apply to all equipment intended for use in explosive atmospheres, whether electrical or mechanical, and also to protective systems.

Manufacturers or suppliers (or importers, if the manufacturers are outside the EU) must ensure that their products meet essential health and safety requirements and undergo appropriate conformity procedures.

This usually involves testing and certification by a third-party certification body (known as a notified body) but manufacturers or suppliers can self-certify equipment intended to be used in less hazardous explosive atmospheres. Once certified, the equipment is marked by the 'EX' symbol to identify it as such.

Note: *As from 1st January 2022, only UKCA-issued 'EX' certificates will be accepted in the UK for products that previously required ATEX Notified Body Certification.*

Certification ensures that the equipment or protective system is fit for its intended purpose and that adequate information is supplied with it to ensure that it can be used safely.

UK
CA

The UKCA mark

09

9.10.3 Competence and training

All the above sets of regulations stipulate the need for a degree of competence (skills, knowledge, training and experience) in assessing certain workplace situations. In most cases it will be necessary for the employer to provide employees with adequate information, instruction, training and supervision to enable them to carry out any work task safely and without risk to their health.

Appendix A – Retrieval of orphaned compressed gas cylinders

The cylinder retrieval arrangements in place for the major national companies are listed below.

Parent company	Collection company	Contact
Calor Gas	CylinderCare	www.wastecare.co.uk/services/cylindercare Tel: 0800 091 0000
BP	Synergy Recycling	www.synergy-recycling.co.uk/ Tel: 0800 083 9652
Flogas	In-house collection by own staff	www.flogas.co.uk/gas-bottles/returns Tel: 0800 574 574
BOC	In-house collection by own staff	www.boconline.co.uk/en/index Tel: 0800 111 333

Orphaned cylinders

1. If a cylinder is no longer needed, it should be returned to the local dealer of the company owning the cylinder.
2. Where the original owner of a compressed gas cylinder cannot be identified, contact CylinderCare.
3. Until such time as they are collected, orphaned cylinders should be stored in a safe and secure manner.
4. If it is not known which company owns an LPG cylinder, the table of LPG cylinder fillers should be viewed in the 'who knows who' guidance table (available online).
5. In extreme circumstances, where all attempts to trace the owner of a cylinder have failed, the Local Authority waste disposal sites may offer a disposal service.

Index

Scan the QR code to complete a brief survey, and share your feedback on the course.

INDEX

INDEX

INDEX

SITE SAFETY PLUS

Construction site health and safety D: High risk activities

Each year there are on average 30 construction-related fatalities. There are also thousands of serious injuries each year, which may include injuries sustained during collapse of excavations, contact with underground services, and confined space and work at height activities (such as roof work and the use of access equipment).

This section covers these high risk activities, which are common on construction sites. It contains important information and guidance to help plan, implement and manage safe systems of work.

ISBN 978-1-85751-584-8

9 781857 515848

V10.1/24 eBook